Tellus 2D

User manual

How to set up Tellus 2D, build and calculate a model in its window, and drive it in plain language through MCP from Claude Code or from TellusHive, with two worked case studies, a strutted excavation (static) and a site shaken by an earthquake (dynamic), and the published case studies against which the program is calibrated.

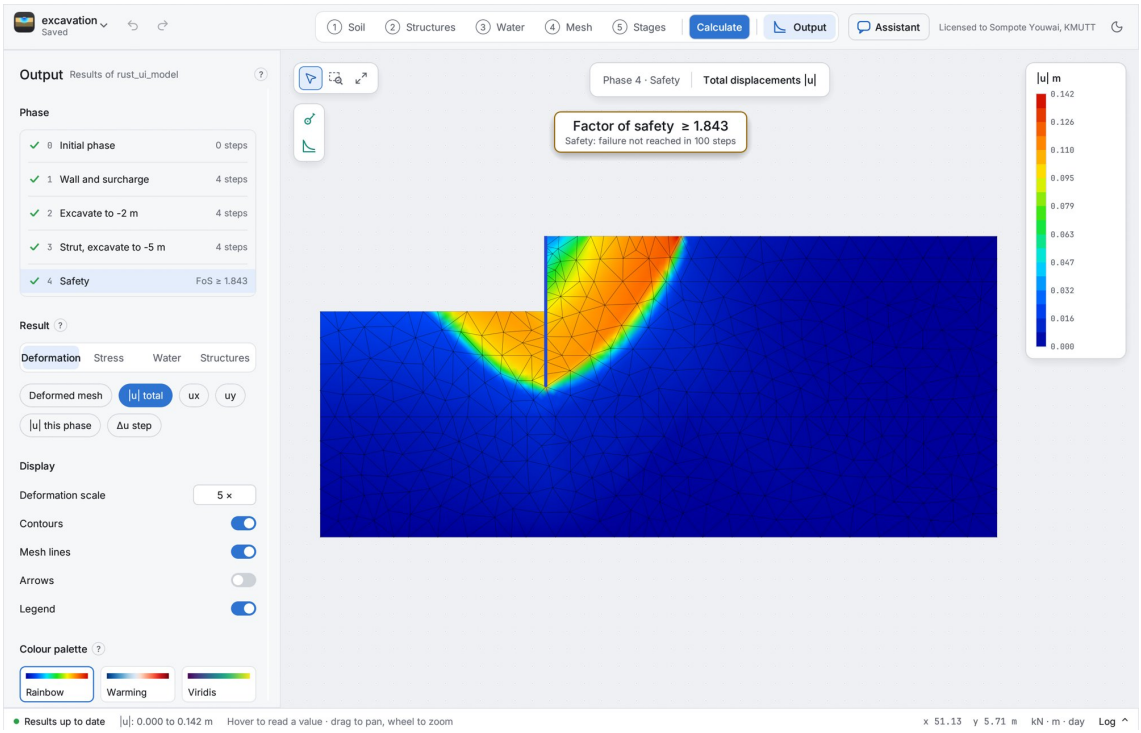


Figure 1. Tellus 2D showing the factor of safety of the strutted excavation of Chapter 4.

Written for Tellus 2D 0.1.4 with the source of 12 November 2026, which adds the updated mesh option. The case study projects are in `manual/cases`.

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1 Introduction

1.1 What Tellus 2D is

Tellus 2D is a plane-strain finite element program for geotechnical engineering. It calculates deformations, stresses, pore pressures, structural forces and factors of safety for problems such as footings, embankments on soft ground, excavations with walls and struts, slopes, and ground shaken by an earthquake. A model is a cross-section through the ground, one metre thick, which suits long structures whose section does not change along their length.

The program can be used in two ways that work on the same project file. In the window you build the model with the mouse in five steps and read the results as coloured pictures and curves. Through MCP an AI assistant such as Claude Code or the agents of TellusHive builds, calculates and reports the model from a request written in plain language, and the project it saves opens in the window for checking. This manual covers both ways and shows the same static case study in each, so you can compare them.

1.2 What it can model

Part	What is available
Elements	15-node triangles, and 25-node rectangles on a structured grid
Soil models	Linear elastic, Mohr-Coulomb, Hardening Soil, HS small, Soft Soil, Soft Soil Creep, hypoplastic sand, and PM4Silt in dynamic phases
Drainage	Drained, Undrained (A), (B) and (C), non-porous
Structures	Plates with interfaces (elastic or elastoplastic), node-to-node anchors, spring supports (struts), geogrids, embedded pile rows, line and point loads, fixities
Water	Phreatic levels, measured pore pressure profiles, steady-state flow, drainage of each boundary, vacuum preloading
Phases	K0 procedure or gravity loading, then Plastic, Consolidation, Safety (phi-c reduction) and Dynamic phases that can branch
Large displacements	Updated mesh option for plastic and consolidation phases
Dynamics	Newmark time integration, Rayleigh damping, viscous, free-field, tied and compliant-base boundaries, earthquake records and dynamic loads
Results	Contours, deformed mesh, plate forces, curves, time histories, response spectra, PDF report, PLAXIS command export

1.3 Units and sign conventions

Lengths are in metres, forces in kilonewtons, stresses and pressures in kPa (kN/m²), and time in days, except in dynamic phases where time is in seconds. The x axis points to the right and the y axis upwards, so depths are negative levels. A negative load acts downwards. Stresses are negative in compression, and the settlement shown in curves is the downward movement taken as positive. Structural forces are per metre run of the section, so the force in one strut or pile is the value times its spacing.

1.4 How this manual is organised

Chapter	Content
2	Installing the program, the licence, and the optional assistant and GPU settings
3	A tour of the window: the Home page, the five steps, Calculate and Output
4	Case study 1 (static): a strutted excavation built step by step in the window
5	MCP: connecting Tellus to Claude Code and asking for an analysis in plain language
6	TellusHive: Tellus as one program in a team of design agents
7	Case study 1 again, through MCP, with the commands and the results
8	Case study 2 (dynamic): a footing on sand over clay in an earthquake, in the window and through MCP
9	Published case studies and the calibration of the program: verification tests, PLAXIS tutorial lessons, field cases, embedded piles, dynamics
10	Structures and their parameters: plates, embedded pile rows and geosynthetics, with anchors, struts and interfaces
11	Water and mesh: water levels, pore pressures, steady flow, drainage, element size, refinement, structured meshes
12	Updated mesh, calculation on a cloud GPU, good practice and troubleshooting
Appendix	The MCP tools

2 Installation and setup

2.1 System requirements

Tellus 2D runs on macOS 11 or later (Intel and Apple Silicon, one universal app) and on 64-bit Windows. It needs about 90 MB of disk space and 73 MB of memory when idle, and a calculation adds roughly 85 MB per 1,000 elements, so 4 GB of memory is enough for ordinary models and 8 GB is comfortable for fine meshes. No graphics card is needed, because the solver runs on the processor unless you connect a cloud GPU (Section 12.2).

2.2 Installing

Download the app from the Tellus website (tellusfem.github.io) or from the public repository github.com/Sompote/tellus2D. The apps are not signed by a registered developer, so each system asks once for permission to open them.

1. **macOS.** Open the DMG and drag **Tellus** to Applications. The first time, right-click the app and choose **Open**, then confirm.
2. **Windows.** Unzip the folder anywhere, for example C:\Tools\Tellus, and start `Tellus.exe`. If Windows shows a warning, choose **More info** and then **Run anyway**. Keep `tigrim.exe` in the same folder, because it is the engine of the assistant.
3. **From source** (developers). With Rust installed, `cargo build --release` builds the app and its agent, and `cargo run -p tellus --release` starts it.

2.3 Entering the licence key at the first start

Tellus runs on a licence key such as `TLS-7KQ4-M2XD-9HPA-R3VW-5NCE`, which you receive with your subscription. The key is limited to a number of computers (seats) and may have an expiry date. You enter it once on each computer, in the window that Tellus shows when it starts without a licence (Figure 2). Until a key is accepted the program shows only this window, so the key is the first thing you type into the software.

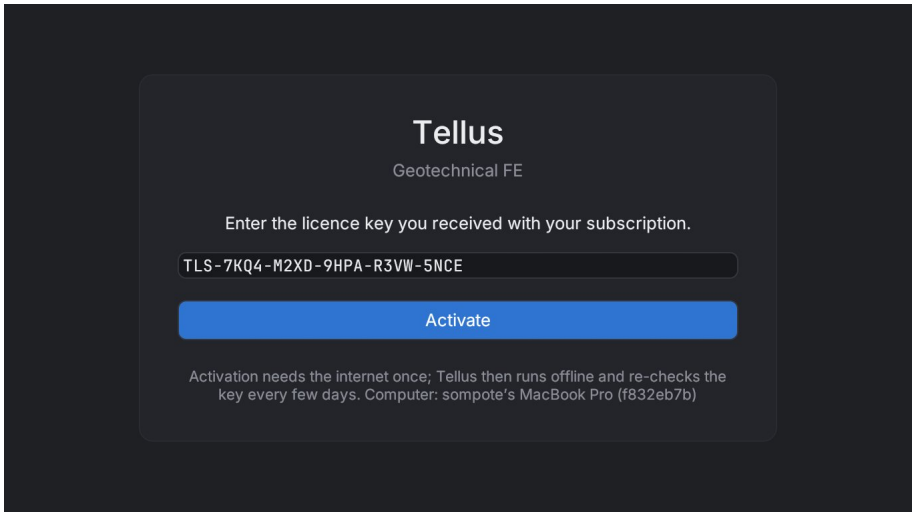


Figure 2. The activation window at the first start: the key field, the Activate button, and the name and id of this computer.

1. Start Tellus. The activation window opens by itself, with the text “Enter the licence key you received with your subscription”.

2. Click in the key field and type or paste the key, in the form `TLS-XXXX-XXXX-XXXX-XXXX-XXXX`. Capital letters and the dashes are part of the key, and pasting avoids typing mistakes.
3. Make sure the computer is connected to the internet, then press **Activate** or the Enter key. The button reads **Checking...** while Tellus contacts the key server.
4. When the key is accepted the window closes and the Home page opens. The top bar now shows **Licensed to** and your name.
5. If the key is refused, a red message under the field says why. Correct the cause from the table below and press **Activate** again.

Message or symptom	Cause	What to do
The Activate button stays grey	Fewer than 8 characters are in the field	Paste the whole key
The key is not known	A typing mistake, or a key for another product	Paste the key again from the e-mail you received
All seats of this licence are in use	The key is already active on its full number of computers	Deactivate Tellus on a computer you no longer use (below), or ask for more seats
The licence has expired or was revoked	The subscription ended	Renew the subscription, then activate again
Cannot reach the licence server	No internet, or a firewall or proxy blocks the server	Check the connection, try another network, or ask your IT department to allow the address of the key server

Activation needs the internet once. Tellus sends the key and a hashed id of the computer to the key server, receives a signed token, and from then on checks the token offline at every start, so it works without a network for up to 30 days and contacts the server again about once a week. The line at the bottom of the activation window shows the name and the id of the computer, which is what the administrator of your licence sees in the list of active seats.

Without a key you can choose **Continue with the free academic edition** on the same window in the released app. The free edition is limited to 1,000 elements, the initial phase and two more phases, the linear elastic, Mohr-Coulomb and Soft Soil models, and two plates, geogrids or springs, it has no embedded piles and no PLAXIS export, and it marks its reports and plots with a watermark. Its badge in the top bar opens the key field again when you receive a key later.

2.4 Checking, moving and freeing the licence

Click **Licensed to ...** in the top bar to open the licence card. It shows the holder, the key, the plan, the end of the subscription and the date until which Tellus runs offline, and it has two buttons. **Check now** contacts the server at once, which is useful after a renewal. **Deactivate this computer** frees the seat, so the key can be used on another computer, and Tellus returns to the activation window. Deactivate before you replace or reinstall a computer, because a seat that is not freed stays counted until the administrator releases it.

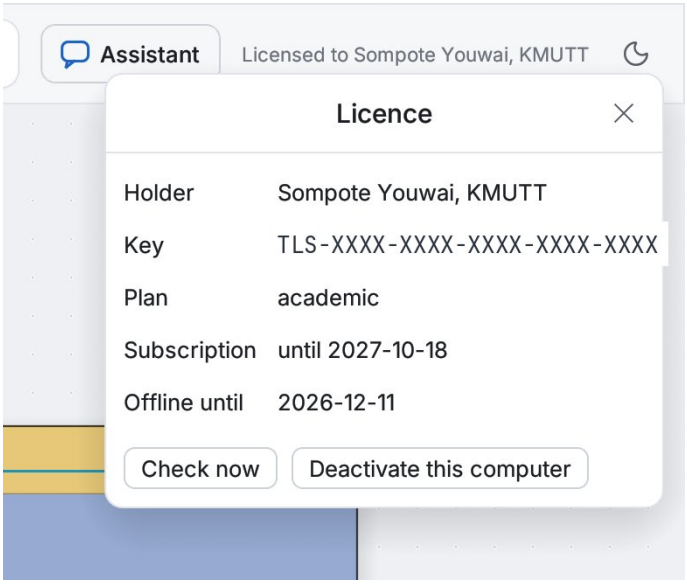


Figure 3. The licence card, opened from the top bar: holder, key (masked here), plan, dates, Check now and Deactivate this computer.

The licence can also be managed from a terminal, which is how you activate a computer that has no screen, such as a server that runs `tellus --mcp`.

```
tellus --activate TLS-XXXX-XXXX-XXXX-XXXX-XXXX   # use a seat on this computer
tellus --licence                                  # the holder, the plan and the dates
tellus --deactivate                               # free the seat before retiring a computer
```

On macOS the program inside the app is `/Applications/Tellus.app/Contents/MacOS/tellus`, and on Windows it is `Tellus.exe` in the folder you unzipped.

2.5 The assistant and its API key

The **Assistant** button at the top right opens a chat panel beside the model, so the model stays in view while you talk to it. You describe the problem in words, in English or Thai, and Tellus builds the model, calculates it and answers with the solver's numbers. Every change goes into the model on screen as an ordinary edit that you can check and undo. The assistant is optional: everything it does can be done by hand in the five steps. It needs an API key of an AI service, which you enter once in the panel's settings (Figure 4).

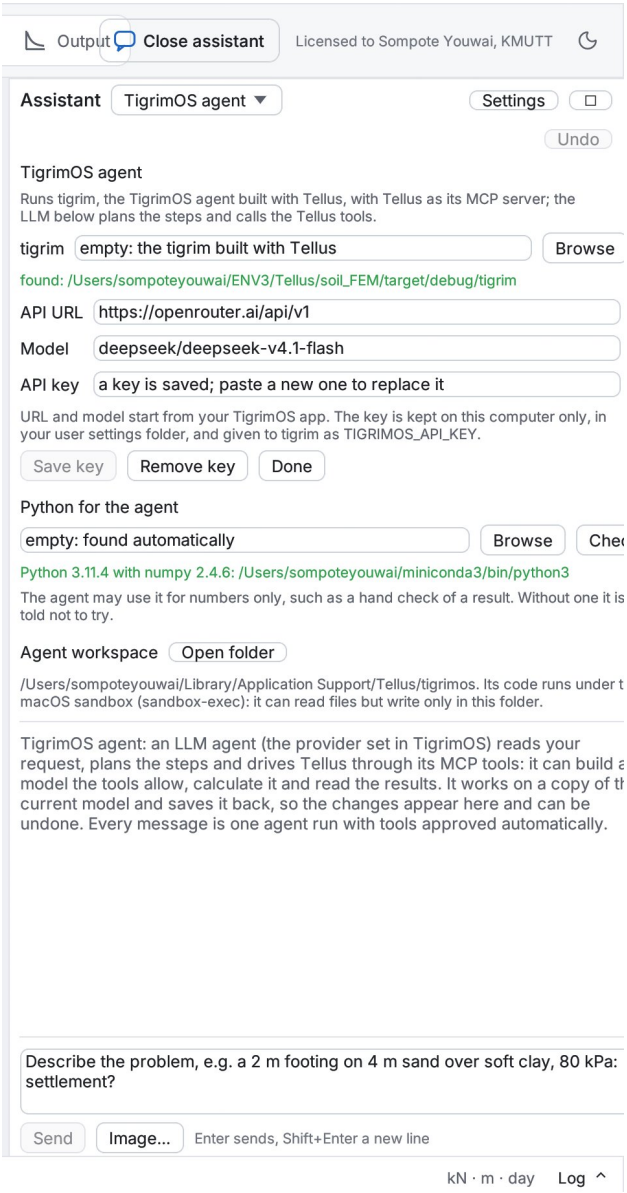


Figure 4. The assistant panel with its settings open for the TigrimOS agent: the engine, the API URL, the model, the API key field, Python for the agent and the agent's workspace, with the message box at the bottom.

Part of the panel	What it does
Engine dropdown, next to Assistant	Chooses Jev or the TigrimOS agent
Settings	Opens and closes the key and options of the selected engine
Undo	Undoes the last change to the model, also one that the assistant made
Clear	Empties the conversation and starts a new one (it appears once there are messages)
Conversation	Your messages, the tool calls of the agent while it works, its replies, pictures of results, and buttons to open a report it wrote
Message box	Where you type. Enter sends and Shift+Enter starts a new line
Send, Stop	Send starts a run. Stop appears during a run and ends it
Image...	Attaches a sketch, a section, a borehole log or a photo, when the provider is OpenRouter and the model takes images
Close assistant, in the top bar	Hides the panel. The conversation is kept

The two engines differ in what they need and what they can do.

	Jev	TigrimOS agent
How it works	Tellus code decides the steps from typed judgments of the Jev model	A language model plans its own steps and uses the Tellus MCP tools
What it can do	A fixed set of actions (layers, a load, a calculation, results)	Anything the MCP tools allow: walls, phases, water, flow, dynamics
Typical time	A few seconds plus the calculation	About 15 s for a simple change plus the calculations
Key you need	A Jev API key from TypeSafe (typesafe.ai)	An API key of an LLM provider, with its API URL and a model name

Setting the API key of the TigrimOS agent. The agent talks to a language model through a provider. Any provider with an OpenAI-compatible API works, as does Anthropic, and you need three things from it: the address of its API, the name of a model, and a key.

1. Create an API key in your account at the provider, and copy it. A key is a long secret text, shown once.
2. In Tellus click **Assistant**, choose **TigrimOS agent** in the dropdown, and click **Settings**.
3. Leave **tigrim** empty. The green line under it shows the agent program that comes with Tellus. Use **Browse** only to point to another build of it.
4. Type the **API URL** of the provider and the **Model** name exactly as the provider writes it (see the table below).
5. Click in **API key**, paste the key, and press **Save key**. The field then reads “a key is saved; paste a new one to replace it”, and the key itself is never shown again.
6. Press **Done**. The settings close and the message box is ready.

Provider	API URL	Model name, examples
OpenRouter (many models with one key)	https://openrouter.ai/api/v1	deepseek/deepseek-v4.1-flash, or any model id from its list
Anthropic	https://api.anthropic.com/v1	claude-opus-5
Another OpenAI-compatible service	The base URL from its documentation, ending in /v1	The model id from its documentation

Setting the API key of Jev. Choose **Jev** in the dropdown and click **Settings**. Paste your Jev API key under **Jev API key**, press **Save key** and then **Done**. Jev needs no URL and no model name.

Both keys are kept on your computer only, in the Tellus settings folder, and are sent only to the service they belong to. **Remove key** deletes a saved key, and pasting a new key and pressing **Save key** replaces it. Treat an API key like a password: do not put it in a project file, a screenshot or an e-mail, and revoke it at the provider if it leaks. The provider charges the use of the key to your account, so a long agent run with many calculations costs more than a short one.

Other settings of the agent. Python for the agent is optional. Leave it empty and Tellus finds Python by itself, and **Check** shows the version it found. The agent uses it for numbers only, such as a hand check of a result. **Agent workspace** is the folder in which the agent works, with **Open folder** to look into it. On macOS the agent's code runs in a sandbox that can read files and write only in that folder.

Using it. Type the request and press Enter, for example “a 2 m footing on 4 m of sand over soft clay, 80 kPa: settlement?”. While the agent works the panel lists its tool calls, and when it has calculated, Tellus calculates the model again in the window and opens the results. Follow up in the same conversation, for example “now lower the water table to 3 m”, and use **Undo** if a change was not what you wanted.

What you see	Cause	What to do
Send stays grey with an orange hint beside it	The key, the URL, the model or the agent program is missing	Open Settings and fill in what the hint names
“tigrim not found” under the tigrim field	The agent program is not beside Tellus	Keep tigrim in the same folder as Tellus, or Browse to it
The run ends with an authentication error	A wrong or revoked key, or a key of another provider	Paste the key again and check that the URL belongs to the same provider
The run ends with “model not found”	The model name is not written as the provider expects	Copy the model id from the provider's list
Image… is grey	The provider is not OpenRouter, or the model does not take images	Hover over the button to read the reason, or choose a model with image input

The assistant can be wrong. It reports numbers that come from the solver, but the choice of soil parameters, model size and phases is still yours to check before a result is used.

2.6 Where Tellus keeps its files

File	Where	Content
Project (.tellus)	Where you save it	The whole model as readable JSON, a few kB. Results are recalculated, not stored
Settings folder	macOS: ~/Library/Application Support/Tellus; Windows: %APPDATA%\Tellus	Licence token, assistant keys and options, view settings, the GPU server address
Report (.pdf)	Next to the project	Written by Output or by the assistant

3 The window

3.1 The Home page

Tellus opens on the Home page (Figure 5), which starts a model in one of three ways. **Soil profile** starts from a borehole whose layers fill the model, which suits level ground with structures in it, and soil clusters can be drawn on top later for embankments and excavations. **Draw geometry** starts from an empty outline in which you draw the soil as polygons, which suits slopes and irregular ground. The text box under them sends a description to the assistant, which builds the model for you. The **Recent** list reopens a project, and a .tellus file dropped anywhere on the page opens too.

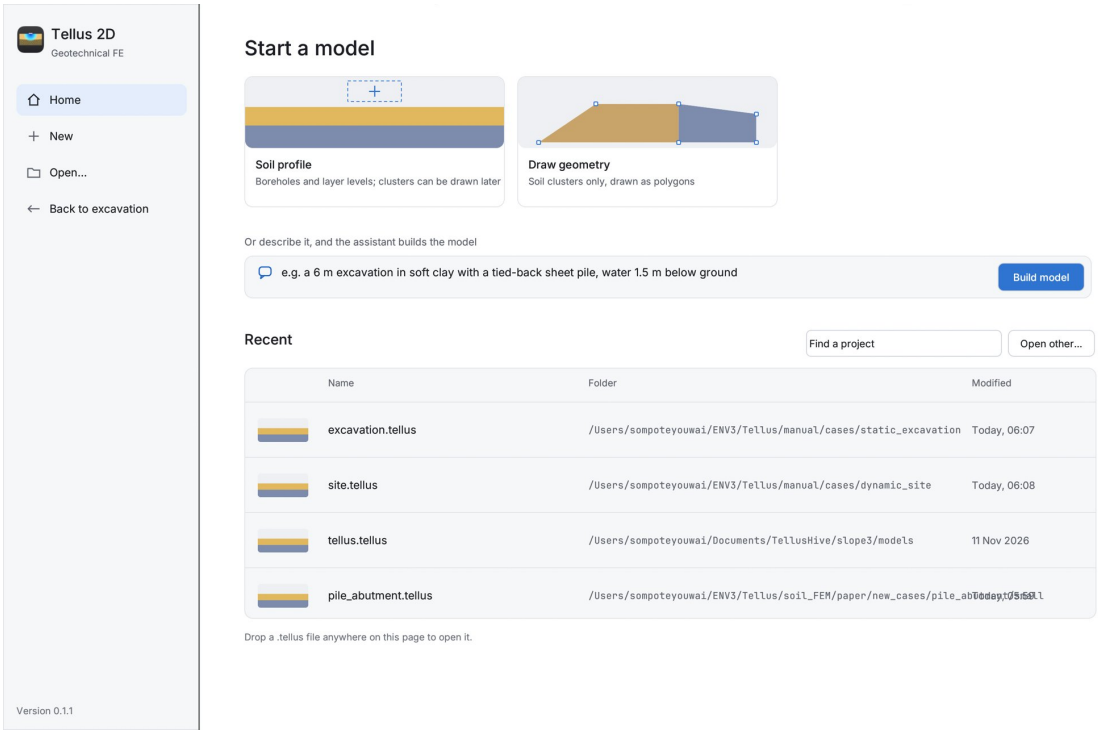


Figure 5. The Home page: two ways to start a model, a description for the assistant, and the recent projects.

3.2 The top bar and the five steps

A model is built in five numbered steps across the top of the window, **Soil**, **Structures**, **Water**, **Mesh** and **Stages**, after which **Calculate** solves the phases and **Output** shows the results. The steps can be visited in any order, because each one edits the same model, and the **Next** button at the bottom of the left panel leads to the following step. The project name at the top left opens the file menu (New, Open, Save, Save as, Home, Export PLAXIS command file), the two arrows next to it undo and redo any edit, and the moon at the far right switches between the light and the dark theme.

Step	What you define there	Left panel
1 Soil	The width of the model, boreholes and layer levels, soil clusters, soil materials	Model extent, Layers, Clusters, Materials
2 Structures	Supports of the sides and base, plates, anchors, springs, piles, geogrids, loads, fixities	Boundaries, Objects
3 Water	Water levels, the unit weight of water, and which boundaries drain	Pore pressure, Flow and drainage

Step	What you define there	Left panel
4 Mesh	Element size, triangles or a structured grid, local refinement, mesh quality	Element size, Refinement, Last mesh
5 Stages	The phases: their type, what soil and structures are in place, the loads and the water	Phase list, phase settings, In this phase

3.3 The drawing area

The drawing area shows the cross-section. The small toolbar at its top left has the pointer, pan, zoom-to-window and fit-to-window tools and the **Snap** distance to which drawn points are rounded. The vertical toolbar holds the drawing tools, for a soil polygon or rectangle, a plate, an anchor, an embedded pile, a geogrid, a spring, a fixity, a line load, a point load, a water level and a mesh refinement. Click an object to select it, drag a handle to move a point, press Delete to remove the selection, and right-click for more commands. The mouse wheel zooms and dragging with the pan tool moves the view. The bar at the bottom shows the state (Ready, Calculating, Results up to date), a hint for the current tool, the units, and **Log**, which opens the solver's messages.

3.4 Materials

Soil and structure materials are edited in one dialog (Figure 6), opened by clicking a material in the Soil step. The list on the left holds the soils and the plate, anchor, pile and geogrid materials. For a soil, the tabs hold its unit weights and model (**General**), stiffness and strength (**Parameters**), permeability (**Groundwater**), interface strength (**Interfaces**), K0 and preconsolidation (**Initial**) and damping (**Dynamics**). A change applies to every layer and cluster that uses the material.

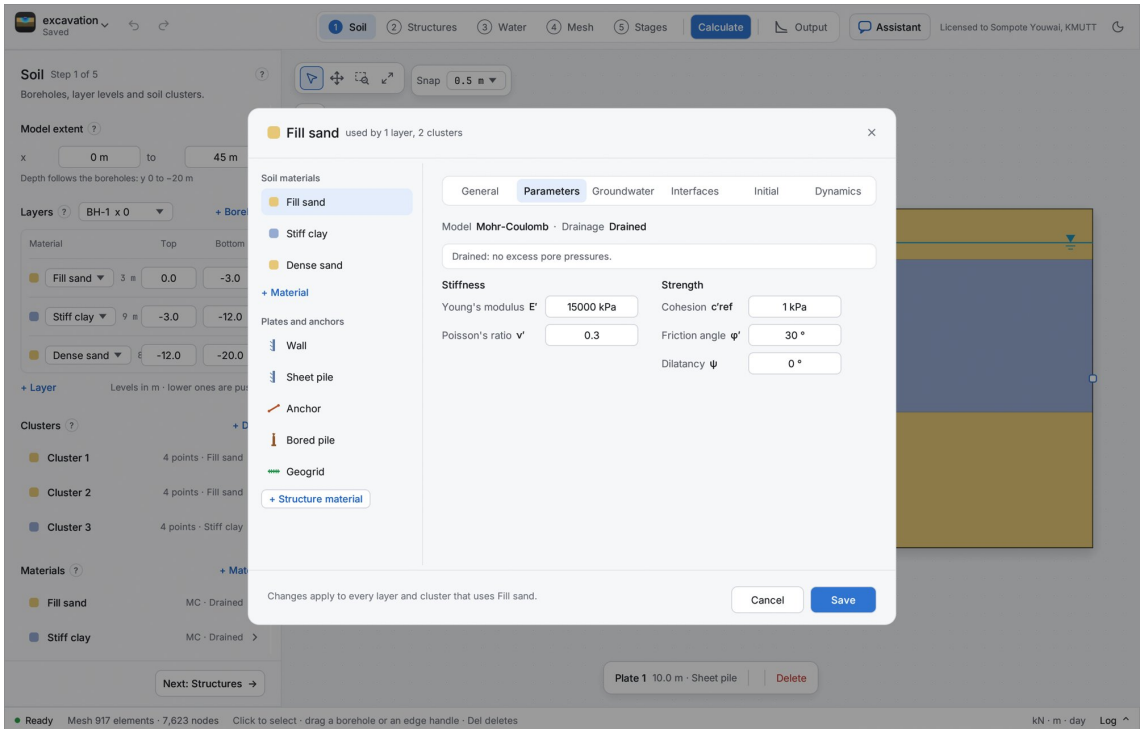


Figure 6. The material dialog on the Parameters tab of a Mohr-Coulomb soil.

3.5 Calculate and Output

Calculate meshes the model if needed and solves the ticked phases in order, while a card shows the phase, the step, the iterations and the plastic points, and **Stop after step** ends the run and keeps the finished phases. **Output** then shows the results of the phase you pick in its list. The **Result** tabs choose the quantity, grouped as Deformation, Stress, Water, Structures and, after a dynamic phase, Dynamics. The **Display** switches set the deformation scale, contours, mesh lines, arrows and legend, and the colour palette can be Rainbow, Warming or Viridis. Hover over the picture to read a value, and use the curve tool at the left of the picture to pick points for curves.

4 Case study 1 (static): a strutted excavation in the window

4.1 The problem

A pit 15 m wide and 5 m deep is dug in front of a sheet-pile wall 10 m long. The ground is 3 m of fill sand over 9 m of stiff clay over dense sand, the water table is 2 m below the surface, and a surcharge of 20 kPa acts on a strip from 3 to 9 m behind the wall. One level of struts at 1 m depth is installed after the first 2 m of digging, and the pit is then dewatered to 0.5 m below its floor. We want the wall deflection, the bending moment, the strut force, the settlement behind the wall and the factor of safety. The model is 45 m wide, so the boundary behind the wall is 30 m away, six times the excavation depth, which is more than the three times that keeps the boundary from affecting the result.

Soil	Level [m]	γ unsat / sat [kN/m³]	E' [kPa]	ν'	c' [kPa]	φ' [°]	ψ [°]	R inter
Fill sand	0 to -3	17 / 19	15,000	0.3	1	30	0	0.67
Stiff clay	-3 to -12	18 / 18	25,000	0.3	10	26	0	0.67
Dense sand	-12 to -20	18 / 20	60,000	0.3	1	36	4	0.67

All three soils use the Mohr-Coulomb model and are drained, which is a simple choice that keeps the example short. For a real stiff clay you would decide between a short-term undrained analysis and a long-term drained one, and a stress-dependent model such as Hardening Soil gives better wall deflections.

4.2 Step 1: Soil

1. On the Home page click **Soil profile**. The model opens with two layers and a strip load.
2. Under **Model extent** set x from 0 to 45 m.
3. Under **Layers** press **+ Layer** once and set the bottoms of the three layers to -3, -12 and -20. The depth of the model follows the layers.
4. Click each material under **Materials**, rename it, and enter the values of the table above on the General, Parameters and Interfaces tabs. Use **+ Material** for the third soil and choose it for the third layer.
5. The soil that will be dug out must be its own cluster. Choose the rectangle tool and draw three rectangles in front of the wall: (0, 0) to (15, -2), (0, -2) to (15, -3) and (0, -3) to (15, -5). Give the first two the fill sand and the third the stiff clay, which you choose in the cluster's card (Figure 8).

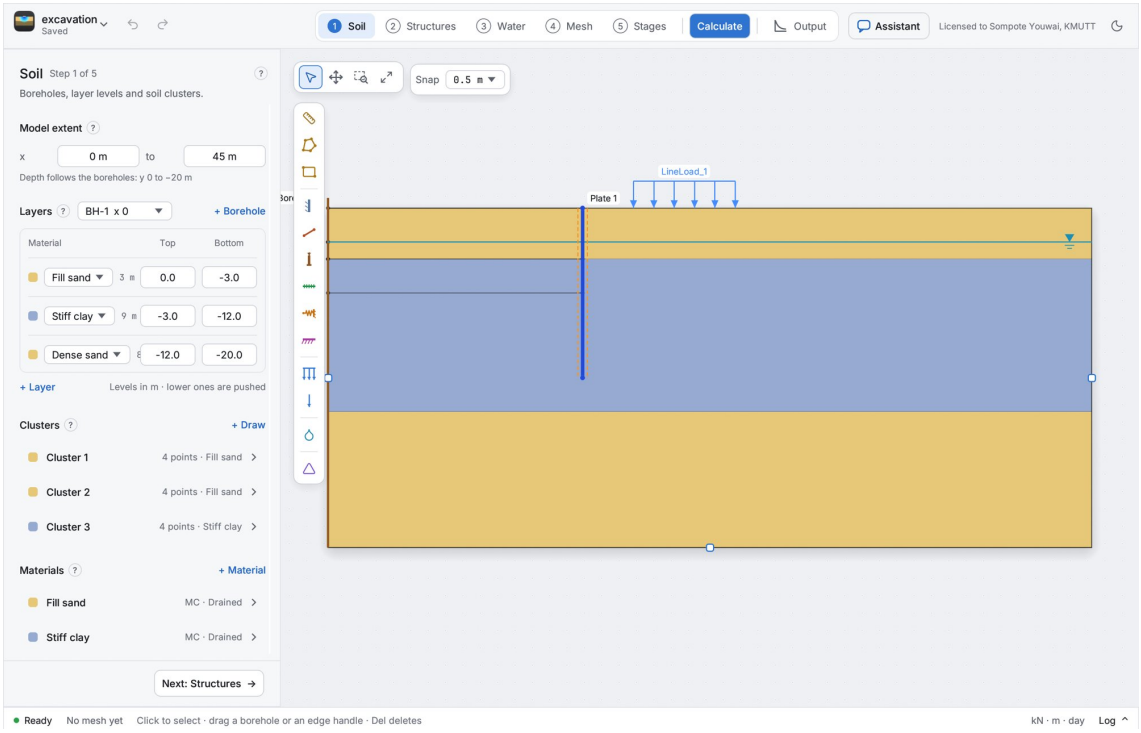


Figure 7. Step 1 Soil: the model extent, the three layers and the three clusters of the pit.

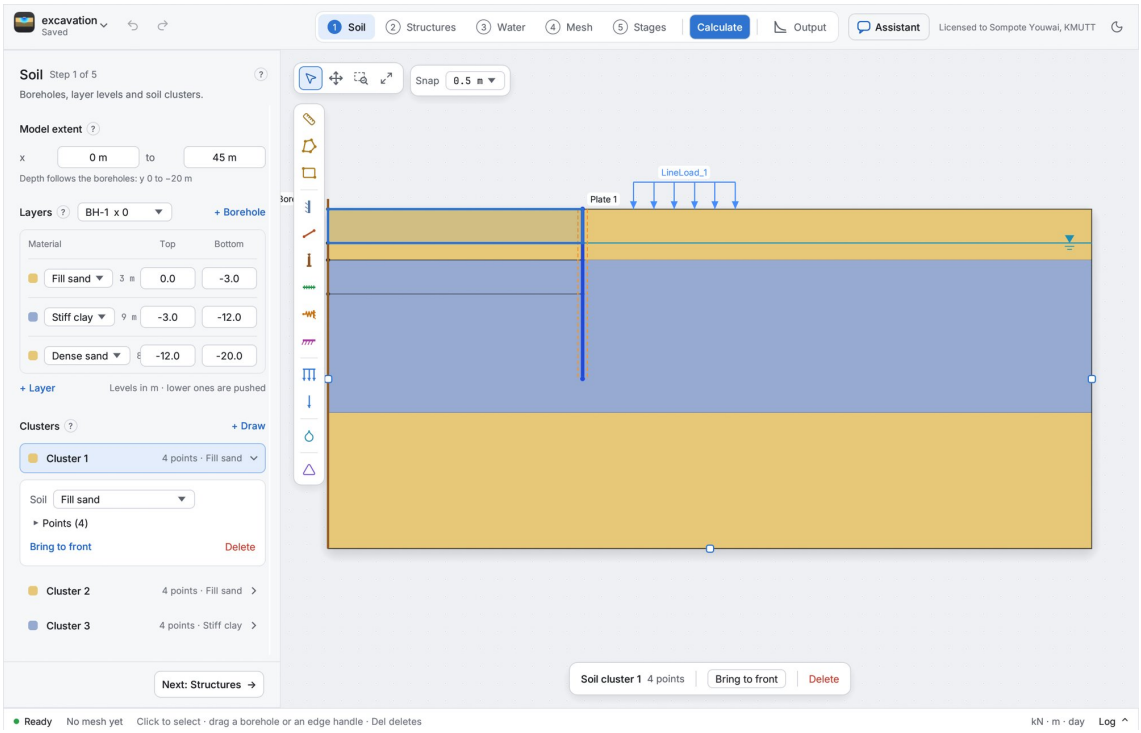


Figure 8. A selected cluster: its soil, its points, and Bring to front, which decides which cluster wins where two overlap.

4.3 Step 2: Structures

1. Leave **Boundaries** at Sides Normally fixed and Base Fully fixed, the usual supports.
2. Choose the plate tool, click at (15, 0) and at (15, -10). In the plate's card choose the material **Sheet pile**, press **Edit** and set EA = 3,000,000 kN/m, EI = 50,000 kNm²/m and w = 1.2 kN/m/m. Keep both **Interfaces** ticked, so the soil can slide along the wall with the strength R inter times that of the soil.

3. Choose the spring tool and click on the wall at (15, -1). Set the angle to 180°, which points from the wall into the pit towards the fixed end, and k = 50,000 kN/m/m. This is EA / (L × spacing) of the strut.
4. Select the line load, move it to x = 18 to 24 m on the surface, and keep it as the surcharge. Its value is set per phase in Stages.

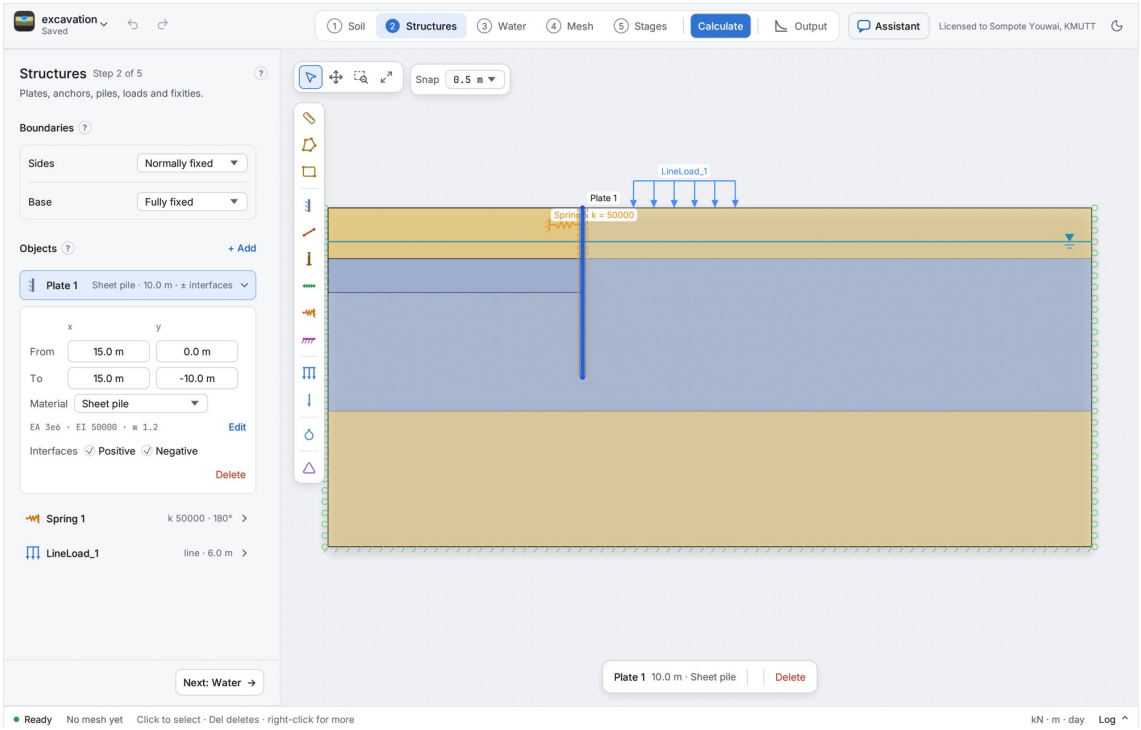


Figure 9. Step 2 Structures: the wall with its interfaces, the strut as a spring, and the surcharge.

4.4 Step 3: Water

1. On **Pore pressure**, set the borehole level to y = -2 m. It is the global level, used by every phase that does not choose another.
2. Press **+ Draw** and draw a second water level through (0, -5.5), (15, -5.5), (15.01, -2) and (45, -2), and name it “Pit dewatered to -5.5”. It lowers the water in the pit and keeps the outside level.

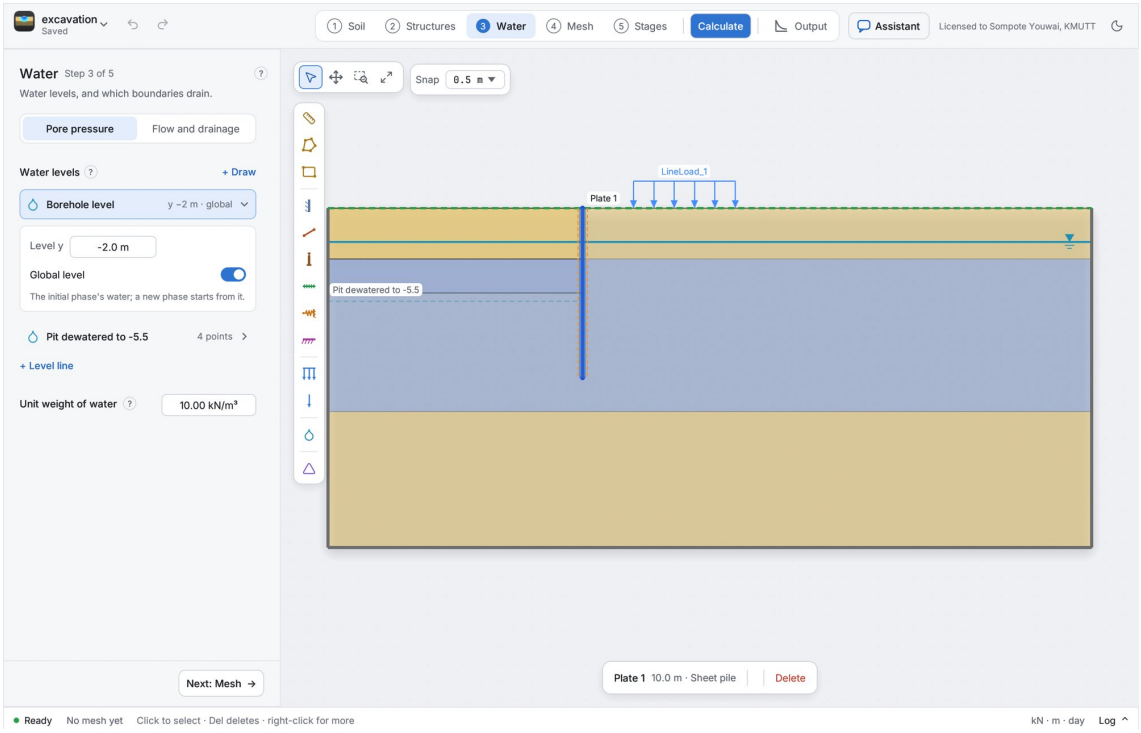


Figure 10. Step 3 Water: the global level and the drawn level of the dewatered pit.

A drawn level gives hydrostatic pressures under each point of the line, which is the simplest model of a dewatered pit. For the flow under the wall, choose **Steady flow** for the phase in Stages and set the draining boundaries on the **Flow and drainage** tab.

4.5 Step 4: Mesh

Choose **Fine** under Element size, or type a relative size, here 0.04 times the model size, which gives elements of about 2 m. Structures are refined by themselves, shown under **Refinement** with their coarseness factor, and **Refine** and **Coarsen** change the element size of the selected layer, cluster, plate or load by a factor of √2. The **Last mesh** card reports the count and the quality. This mesh has 917 elements and 7,623 nodes, with a smallest angle of 25° (Figure 11).

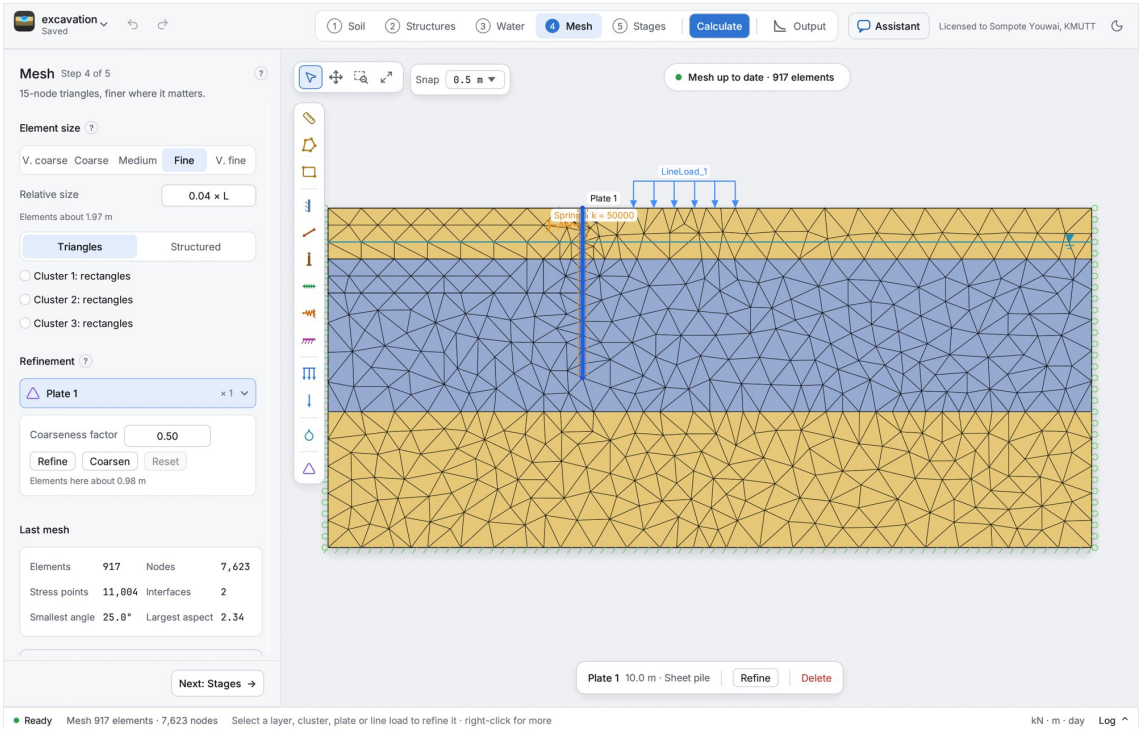


Figure 11. Step 4 Mesh: element size, refinement of the wall, and the quality of the last mesh.

4.6 Step 5: Stages

Every phase starts as a copy of the one before, and you then change what differs. Select a phase in the list to see its settings, and switch soil, structures and loads on or off either with the switches under **In this phase** or by clicking them in the picture. The initial phase calculates the stresses in the ground before construction with the K0 procedure, which is right for level ground.

Phase	Type	Starts from	What changes
0 Initial phase	K0		All soil in place, no structures, global water level
1 Wall and surcharge	Plastic	0	Plate on, line load on at -20 kN/m/m
2 Excavate to -2 m	Plastic	1	Cluster 1 off
3 Strut, excavate to -5 m	Plastic	2	Spring on, clusters 2 and 3 off, water level “Pit dewatered to -5.5”
4 Safety	Safety	3	Nothing: a Safety phase reduces the strength of the state it starts from

1. Select phase 1, name it, and switch on the plate and the line load. Enter -20 for the load.
2. Press **+ Phase**, name it “Excavate to -2 m”, and click cluster 1 in the picture to switch it off.
3. Press **+ Phase** again, switch on the spring, switch off clusters 2 and 3, and choose the dewatered level under **Water**.
4. Press **+ Phase** and set its **Type** to Safety.

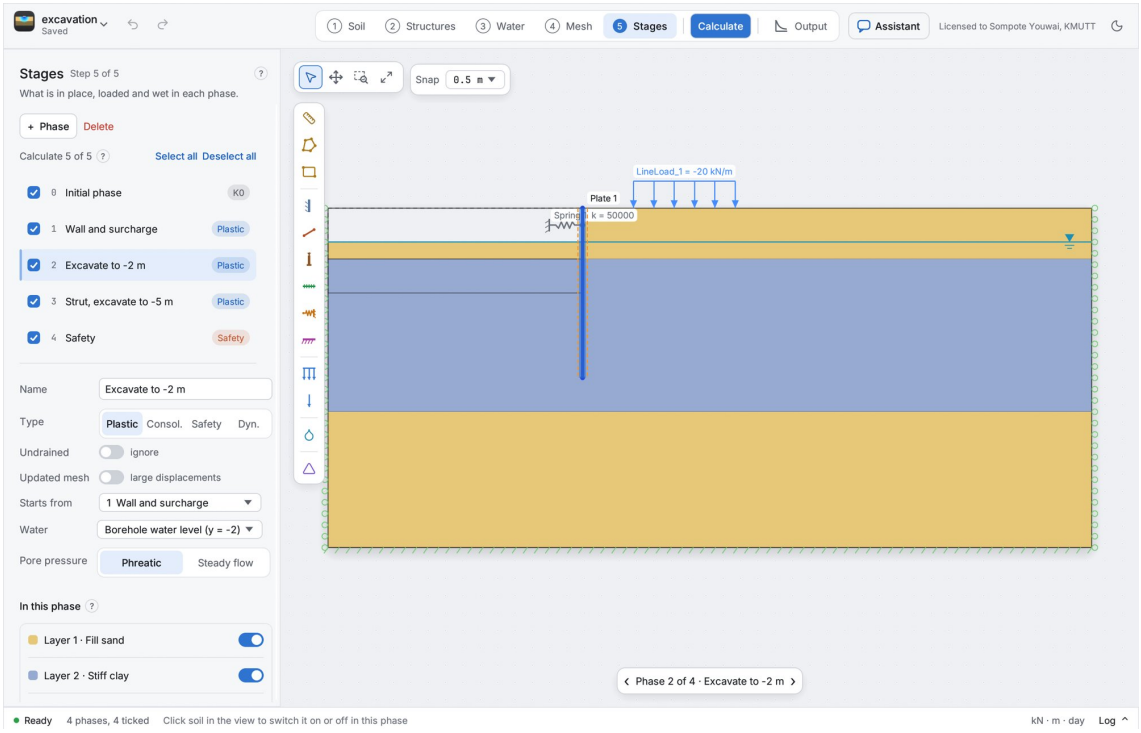


Figure 12. Step 5 Stages: phase 2 with the first cluster dug out. The switches Undrained and Updated mesh belong to the phase.

4.7 Calculate

Press **Calculate**. The card of Figure 13 shows the progress, and the ticks in the phase list show what is done. This model takes about 77 s on a laptop, most of it in the Safety phase, which takes 100 steps.

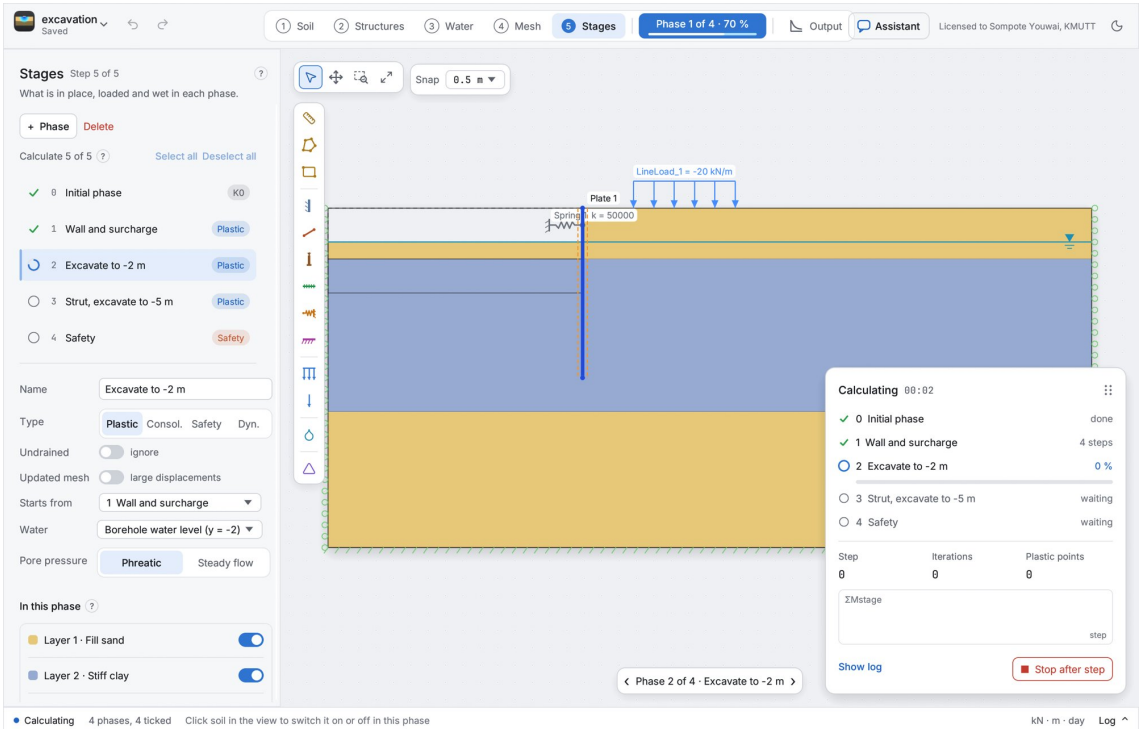


Figure 13. The calculation card: phases done, the current step, iterations and plastic points.

4.8 Results

Output opens on the last phase. Figure 14 shows the total displacements of the Safety phase, which draw the failure mechanism: a wedge behind the wall that slides into the pit and under the wall toe. The label reads factor of safety ≥ 1.84, with the note that failure was not reached in 100 steps, so the true value is at least this number and a longer run would refine it.

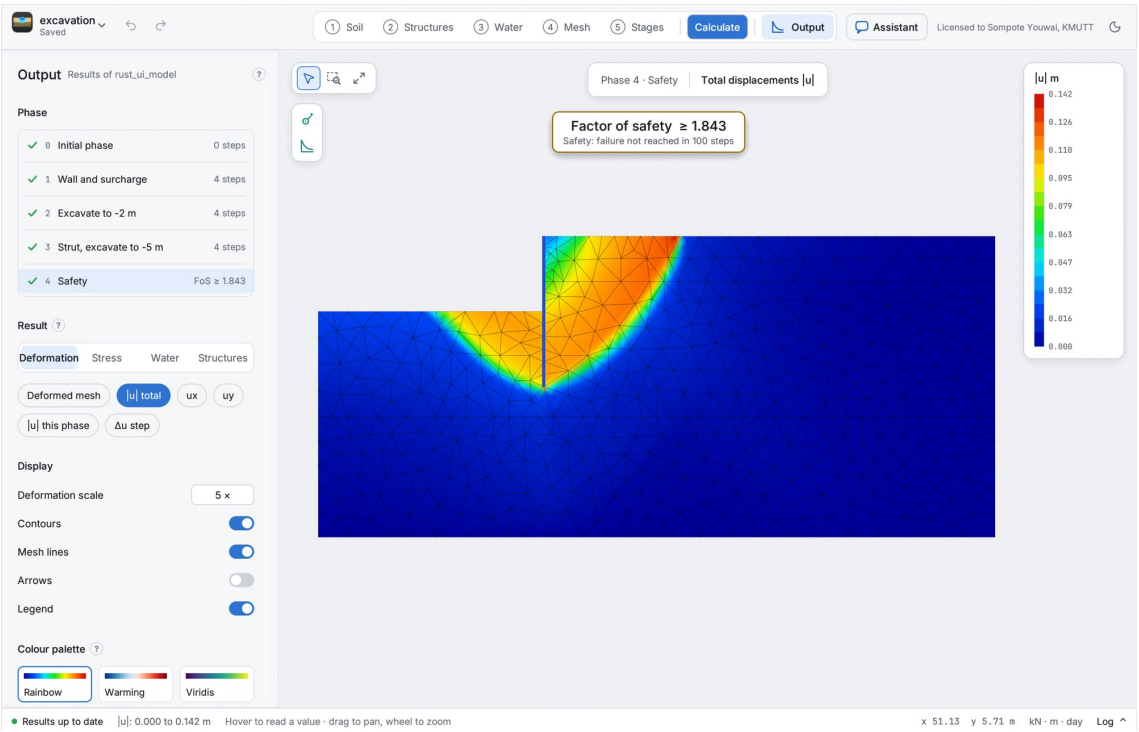


Figure 14. Output: total displacements of the Safety phase and the factor of safety.

Result at the end of excavation (phase 3)	Value
Largest wall movement into the pit	12.2 mm at y = -6.0 m
Largest bending moment in the wall	52.7 kNm/m at y = -3.8 m
Largest shear force in the wall	40.6 kN/m at y = -1.1 m
Strut force (spring)	64.4 kN/m, compression
Largest settlement behind the wall	8.6 mm at 6 m from the wall
Factor of safety (phase 4)	at least 1.84

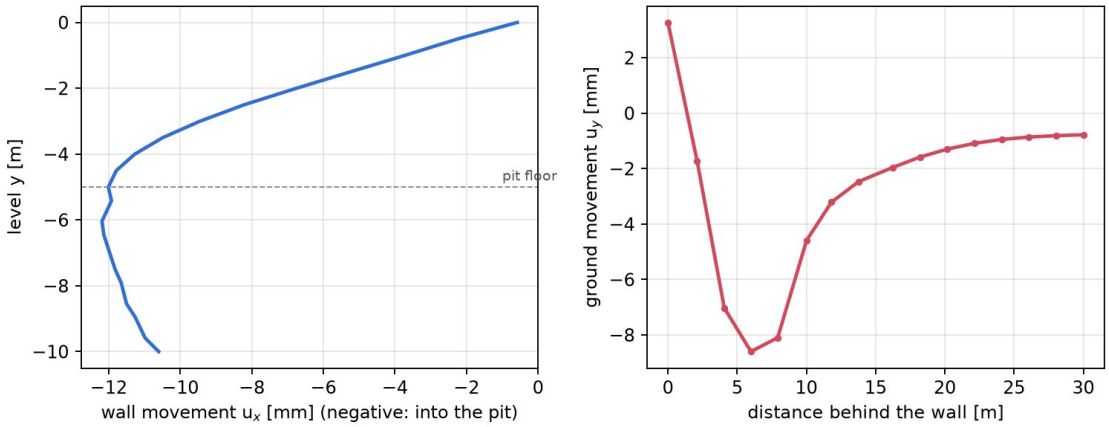


Figure 15. Wall movement with depth and ground movement behind the wall at the end of excavation.

For structural forces choose **Structures** and then Plate N, Plate Q or Plate M, which draws the diagram along the wall with its extreme values (Figure 16). For a curve, click the curve tool, press **+ Pick** and click a point of the model, then choose what the axes show, for example the settlement of a point against the step, or ΣMsf against the displacement to judge whether a Safety phase has reached failure (Figure 17). **Copy data** puts the curve on the clipboard for a spreadsheet.

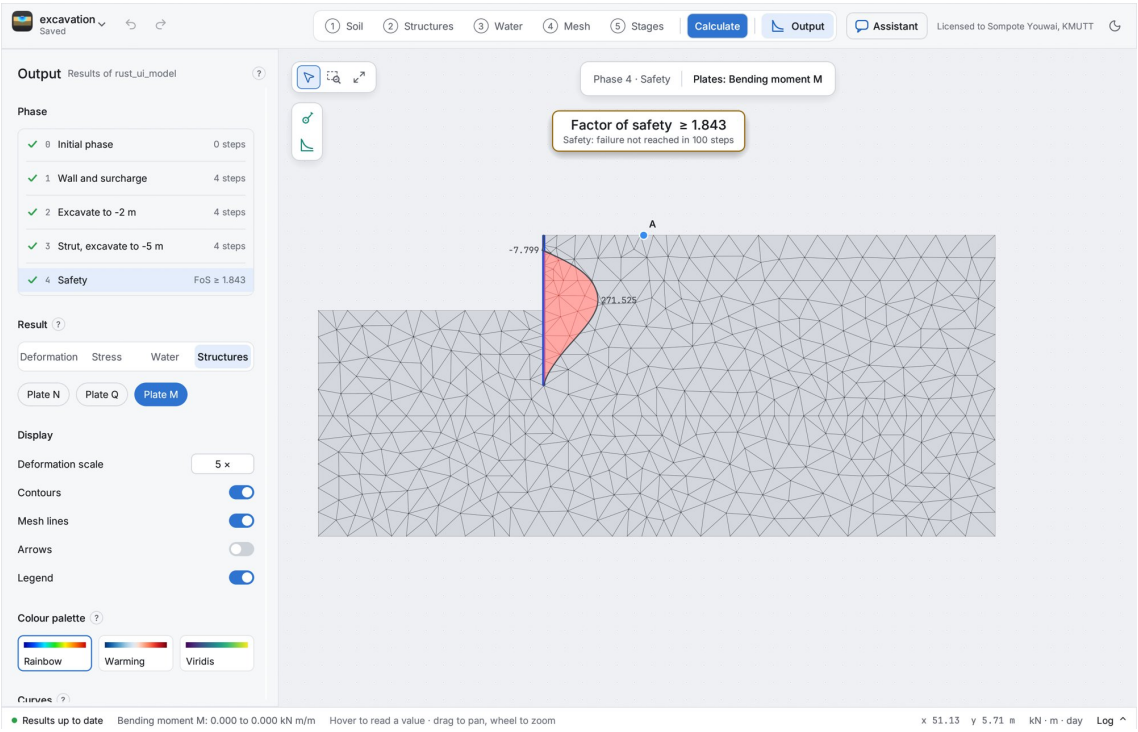


Figure 16. Bending moment along the wall. The picture is of the Safety phase, where the reduced strength raises the moment.

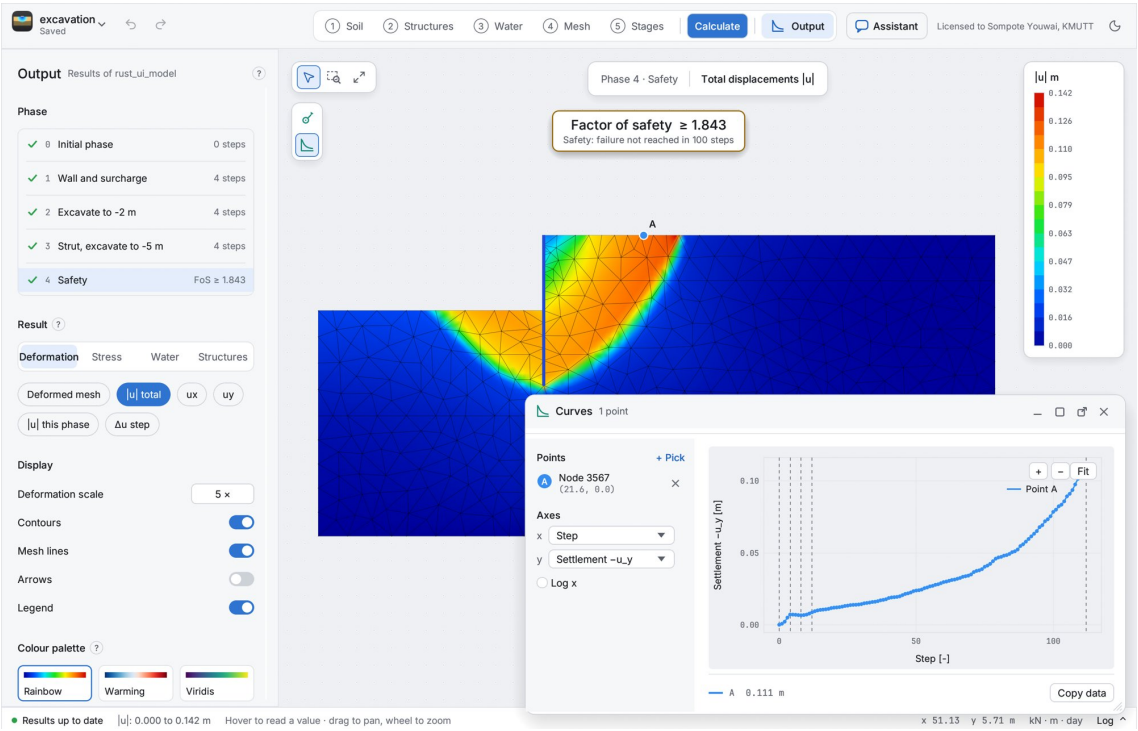


Figure 17. The Curves window: a picked point and its settlement against the calculation step.

Check a result before you use it. The strut force is per metre run, so a strut every 3 m carries three times the value. The wall moves 12 mm with a linear elastic, perfectly plastic soil whose stiffness you chose, and that choice governs the answer more than the mesh does. A factor of safety from a phi-c reduction depends on the mesh by roughly a tenth, so compare it with a hand check when it is close to the required value.

5 Tellus through MCP with Claude Code

5.1 What MCP gives you

MCP, the Model Context Protocol, is a standard way for an AI assistant to use a program's functions as tools. Started as `tellus --mcp`, Tellus runs without a window and offers 37 tools that build a model, calculate it and read the results. The assistant chooses and calls the tools, and Tellus does the engineering calculation, so the numbers in the answer come from the solver and not from the language model. The server works on the same .tellus files as the window, so a project that the assistant saves opens in the window for checking, and a project you built by hand can be handed to the assistant.

5.2 Connecting Tellus to Claude Code

Claude Code is Anthropic's coding assistant for the terminal, the desktop and the IDE. Register Tellus with it once, with the full path of the program:

```
# macOS, installed app
claude mcp add -s user tellus -- /Applications/Tellus.app/Contents/MacOS/tellus --mcp

# Windows
claude mcp add -s user tellus -- "C:\Tools\Tellus\Tellus.exe" --mcp

# a build from source, run in the repository folder
claude mcp add -s user tellus -- "$PWD/target/release/tellus" --mcp
```

1. Activate the licence on the computer first (Section 2.3), because the server checks it before it offers its tools.
2. Run the command above in a terminal. `-s user` makes Tellus available in every folder you work in.
3. Start a new Claude Code session and type `/mcp`. Tellus appears in the list with its tools.
4. Ask for an analysis in plain language. Claude Code asks for permission the first time it uses each tool.

Other MCP clients start the same command as a stdio server. Claude Desktop and Cursor take it under `mcpServers` in their configuration file, and VS Code under `servers` with `"type": "stdio"`:

```
{ "mcpServers": { "tellus": { "command": "/Applications/Tellus.app/Contents/MacOS/tellus",
                              "args": ["--mcp"] } } }
```

To start the server with a project loaded, add `"--open", "/path/to/model.tellus"` to the arguments.

5.3 Asking for an analysis

Write the request as you would brief an engineer, with the geometry, the soil, the water, the loads, the construction sequence and the results you want. The assistant follows the flow that the server describes to it: new project, materials, layers or clusters, structures, phases, validate, calculate, read results. A request that leaves out a value makes the assistant choose one, so say which values are given and ask it to state the ones it assumed.

```
Use Tellus. Model a 2 m wide strip footing on 4 m of medium dense sand (E 30 MPa, phi 32)
over soft clay (undrained, su 25 kPa, Eu 5 MPa) to 12 m depth. Water table at 1.5 m.
Apply 80 kPa on the footing. Give me the settlement under the centre, the factor of
safety, and a picture of the failure mechanism. Save the project as footing.tellus and
```

```
list every parameter you assumed.
```

Useful follow-up requests are “make the model wider and tell me whether the settlement changes”, “refine the mesh under the footing and repeat”, “plot the settlement against the load” and “write a PDF report”. Because the model is saved as a .tellus file, you can open it in the window at any time and look at what the assistant built.

5.4 Rules the assistant follows, and what to check

Topic	What the server tells the assistant	What you should check
Units	m, kN, kPa, days, y upwards, negative load downwards	That a stiffness in MPa was entered as kPa × 1000
Model size	Side boundaries at least 3 times the slope height or excavation depth away, embankment models 3 times the base width	That widening the model does not change the result
Undrained soil	Undrained (B) and (C) use su with phi = 0	Which drainage type each clay was given, and why
Failed phases	A phase that does not converge is reported with the load fraction reached, and later phases are not calculated	Whether a failure is physical or a modelling mistake
Results	Report only numbers that the tools returned	The picture of the deformed mesh, which shows a wrong model at a glance

The assistant is quick at building and varying models, and it can still choose a poor parameter, misread a sketch or describe a result too confidently. Treat its model as a draft by a junior engineer: open it, read the phases and the materials, and decide yourself whether the answer may be used.

6 Tellus in TellusHive

6.1 What TellusHive adds

TellusHive is a design office of AI agents. A coordinator reads your request and gives the work to specialists, for example a geotechnical engineer, a structural engineer, a checker, a drafter and a report writer, and each specialist uses the programs it is allowed to, connected as MCP plugins. Tellus 2D is one of these programs, next to Tellus3D, TellusLEM for slope stability, Pondus for structures, Aedifex for buildings and terrain, and CADCraft for drawings. Use Claude Code when you want to work on one Tellus model yourself in a conversation, and TellusHive when you want a whole design with calculation sheets, drawings and a report that cite the standards you chose.

6.2 Setting it up

1. Install TellusHive from github.com/Sompote/TellusHive (a macOS app and a Windows program) and start it.
2. **Add a model.** Open **Settings** (the gear at the top right) and paste an API key for OpenRouter or DeepSeek. Without a key, give an agent the provider `claude-code` or `codex` in the Team step, and it runs on your Claude Code or Codex login.
3. **Make a project.** On the Home page start from an example, a template or a one-line brief. Step **1 Project** holds the description and the notes that every agent reads.
4. **Choose the standards.** In step **2 Standards** import the codes you design to (PDF, text or Markdown) and tick the ones this project follows, so the agents cite real clauses.
5. **Check the team.** Step **3 Team** shows the coordinator and the specialists. Click an agent to set its model, its thinking effort, its tools and its plugins.
6. **Connect Tellus.** Step **4 Plugins** lists the MCP programs found on your computer. Tellus appears when the app is installed and activated. Tick it for the agent that does the geotechnical analysis.

6.3 Asking for a design

In the **Design office** write the request with all the data and name every deliverable, because the coordinator gives work only to the specialists the request needs. The panel on the left shows which agents are working and what they are doing, and the results appear beside the chat and in **Output**: calculation sheets with PASS or FAIL, drawings, the PDF report, and the Tellus project that the geotechnical agent saved in the project's `models` folder, which you can open in the Tellus window.

```
Design the temporary support of a 5 m deep, 15 m wide excavation: sheet-pile wall 10 m long
with one level of struts at 1 m depth. Ground: 3 m fill sand (phi 30, E 15 MPa), 9 m stiff
clay (c' 10 kPa, phi 26, E 25 MPa), then dense sand. Water at 2 m, pit dewatered to 5.5 m.
Surcharge 20 kPa from 3 to 9 m behind the wall. Use Tellus 2D for the wall deflection,
bending moment, strut force and factor of safety. Check the sheet-pile section and the
strut in steel. Deliverables: calculation sheets, a section drawing and a PDF report.
```

Every number in a TellusHive sheet traces back to a tool result, and the full log of each agent is in the project's `sessions` folder. The design is still yours to check before it is used, as with any assistant.

7 Case study 1 again, through MCP

7.1 The request

The excavation of Chapter 4 can be given to Claude Code in one message. The text below holds the same data as the steps of Chapter 4, and the assistant turns it into the tool calls of the next section.

```
Use Tellus to analyse a strutted excavation and save it as excavation.tellus.
Ground from the surface: 3 m fill sand (gamma 17/19, E 15000 kPa, nu 0.3, c 1, phi 30),
9 m stiff clay (gamma 18, E 25000, nu 0.3, c 10, phi 26), 8 m dense sand (gamma 18/20,
E 60000, nu 0.3, c 1, phi 36, psi 4). All Mohr-Coulomb, drained, Rinter 0.67.
Water level at -2 m. Model 45 m wide. Sheet-pile wall at x = 15 from 0 to -10 m
(EA 3e6 kN/m, EI 5e4 kNm2/m, w 1.2), with interfaces. Surcharge 20 kPa from x = 18 to 24.
Phases: (1) wall and surcharge, (2) excavate x < 15 to -2 m, (3) strut at -1 m
(spring, k 5e4 kN/m/m, towards the pit), excavate to -5 m, pit dewatered to -5.5 m,
(4) safety. Mesh size 0.04. Report the wall deflection, the largest moment, the strut
force, the settlement behind the wall and the factor of safety.
```

7.2 The tool calls

The calls below are in order. They are also in `manual/cases/static_excavation/make_case.py`, a short Python script that starts `tellus --mcp` and sends the same calls, which is the way to run a case again or to vary it in a loop without an assistant.

Tool	Arguments (shortened)	Purpose
new_project	kind: profile	A borehole model with two default soils and a strip load
set_contour	x_min 0, x_max 45, water_level -2	Width and water table
update_soil_material × 2, add_soil_material	name, model, E, nu, c, phi, psi, gamma, drainage, rinter	The three soils
set_soil_layers	top 0, layers [{thickness 3, material 0}, {9, 1}, {8, 2}]	The profile
add_soil_cluster × 3	polygon [[0,0],[15,0],[15,-2],[0,-2]], material 0 …	The soil to dig out
set_initial_phase	method k0, clusters_on [true, true, true]	Initial stresses, all soil in place
add_plate	start [15,0], end [15,-10], interfaces true, material {ea, ei, w}, active_in none	The wall
add_spring	at [15,-1], angle 180, k 50000, active_in none	The strut
add_water_level	points [[0,-5.5],[15,-5.5],[15.01,-2],[45,-2]]	The dewatered pit
update_phase 0	name, plates [true], loads [{index 0, value -20}]	Phase 1
add_phase	clusters_on [false, true, true]	Phase 2
add_phase	springs [true], clusters_on [false, false, false], water_level 0	Phase 3
add_phase	calc safety	Phase 4
generate_mesh, validate, calculate		917 elements, check, solve
get_results, get_plate_forces	quantity ux at points along the wall, phase 3	The answers

Tool	Arguments (shortened)	Purpose
show_picture, write_report, save_project	quantity, phase, file	Pictures, PDF report, project file

7.3 The results

The server returns the same numbers as the window, because both use the same solver: a wall movement of 12.2 mm, a largest moment of 52.7 kNm/m, a strut force of 64.4 kN/m and a factor of safety of at least 1.84. Figure 18 is the picture that `show_picture` writes for the deformed mesh, and the report from `write_report` is `report.pdf` in the case folder.

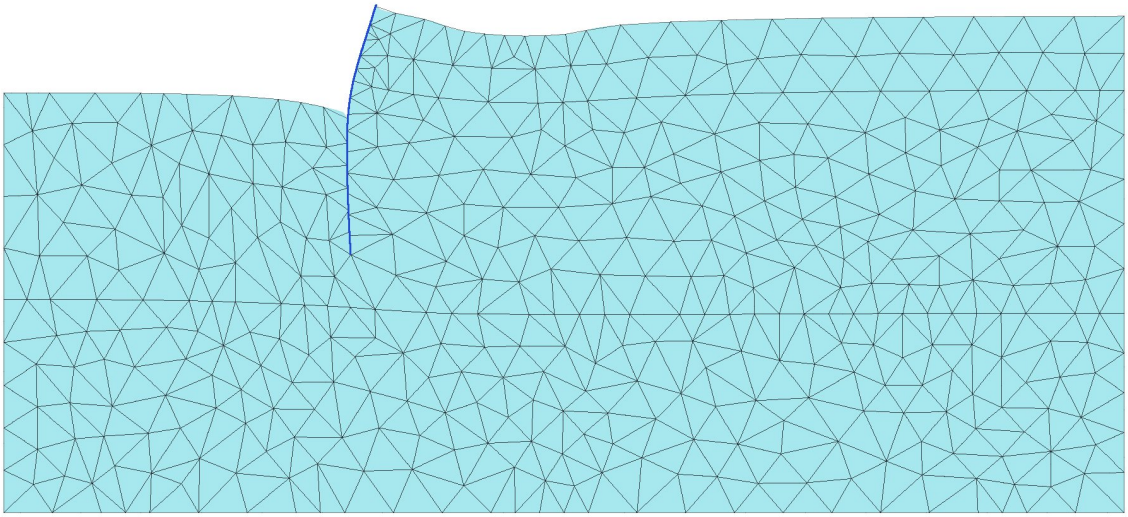


Figure 18. Deformed mesh at the end of excavation, written by show_picture (displacements scaled 100 times).

In a script, each call is one line. The helper `mcp_client.py` in `manual/cases` starts the server and returns each tool's answer as a Python dictionary:

```
from mcp_client import Mcp, values
m = Mcp("solver.log")
m.tool("new_project", {"kind": "profile"})
m.tool("set_contour", {"x_min": 0, "x_max": 45, "water_level": -2.0})
m.tool("add_plate", {"start": [15, 0], "end": [15, -10], "interfaces": True,
                     "material": {"name": "Sheet pile", "ea": 3.0e6, "ei": 5.0e4, "w": 1.2},
                     "active_in": "none"})
...
phases = m.tool("calculate")["phases"]
wall = values(m, 3, "ux", [[15, -0.5 * j] for j in range(21)])     # phase 3, along the wall
print(phases[4]["factor_of_safety"])
```

8 Case study 2 (dynamic): a footing on sand over clay in an earthquake

8.1 The problem

A strip footing 3 m wide carries 80 kPa on 5 m of sand over 15 m of clay. The site is shaken by a recorded earthquake motion, the Gilroy record of the 1989 Loma Prieta earthquake with a peak of 0.357 g, applied at the base of the model as an outcrop motion. We want the acceleration at the surface compared with the input, the movement of the footing during the shaking, and the response spectrum at the surface. The model is 60 m wide and 20 m deep, and the first 8 s of the record are calculated, which contain its strong part.

Soil	Level [m]	γ unsat / sat [kN/m³]	E' [kPa]	ν'	c' [kPa]	φ' [°]	Damping
Sand	0 to -5	17 / 20	60,000	0.3	1	32	5 % at 1 and 10 Hz
Clay	-5 to -20	17 / 17	30,000	0.3	40	0	5 % at 1 and 10 Hz

A dynamic analysis needs three things that a static one does not: a signal, which is a function of time, damping for each soil, and boundaries that let waves leave the model instead of reflecting them. The stiffness that matters is the small-strain stiffness of the soil, which is several times the static value, so derive E from the shear wave velocity when you have one, with $G = \rho V_s^2$ and $E = 2 G (1 + \nu)$.

8.2 The static part

Build the two layers, the materials and the footing load as in Chapter 4: a soil profile from 0 to 60 m with the water level at -2 m, the line load from x = 28.5 to 31.5 m, and one plastic phase “Footing load” with -80 kN/m/m. A dynamic phase starts from the stresses of the phase before it, so the static phases must be right first. Choose a mesh size of 0.05, which gives 411 elements. As a rule, an element should be shorter than about one fifth of the shortest wavelength of interest, which is the shear wave velocity divided by the highest frequency, and the 15-node elements of Tellus have several nodes along each side, which helps.

8.3 The signal

1. In Stages, open **Signals** (Figure 19). A signal is a harmonic function, a Ricker pulse, or a table of time and value.
2. Press **+ Table / record** and **Import file...**. Tellus reads PEER .AT2 files, two columns of time and value, or one value per line with the time step entered beside it.
3. Keep the record in units of g. The phase multiplies it by 9.81 to get m/s². The buttons under **Scale** convert or scale a record that is in other units.
4. Switch on **Drift correction** for an accelerogram, so the velocity and the displacement end at zero.

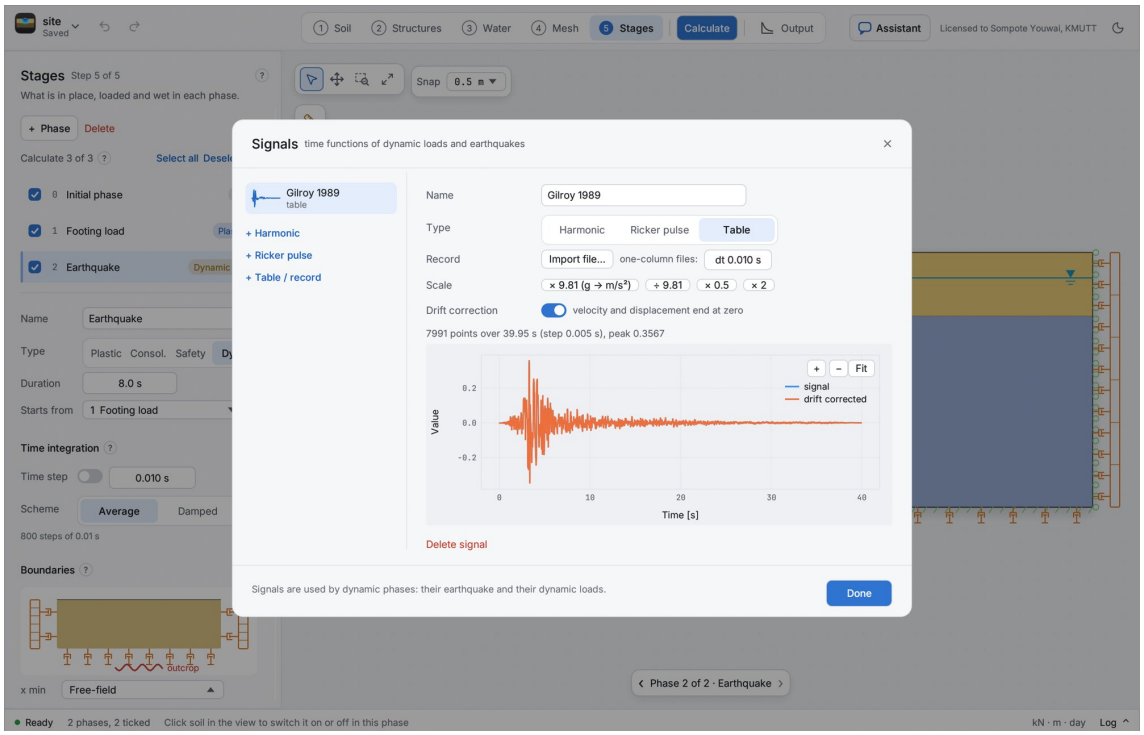


Figure 19. The Signals dialog with the imported record: 7,991 points over 40 s, peak 0.357 g.

8.4 Damping

Open a soil material and its **Dynamics** tab (Figure 20). Rayleigh damping is entered as a damping ratio at two target frequencies, and Tellus shows the resulting curve and the coefficients α and β. Between the two frequencies the damping is lower than the target and outside them it rises, so choose them to bracket the frequencies that matter, often the natural frequency of the deposit and a few times that. Here 5 % at 1 and 10 Hz gives α = 0.571 1/s and β = 0.00145 s. The second tab, PM4Silt, switches a silt or clay to a cyclic model that builds up pore pressure during the shaking.

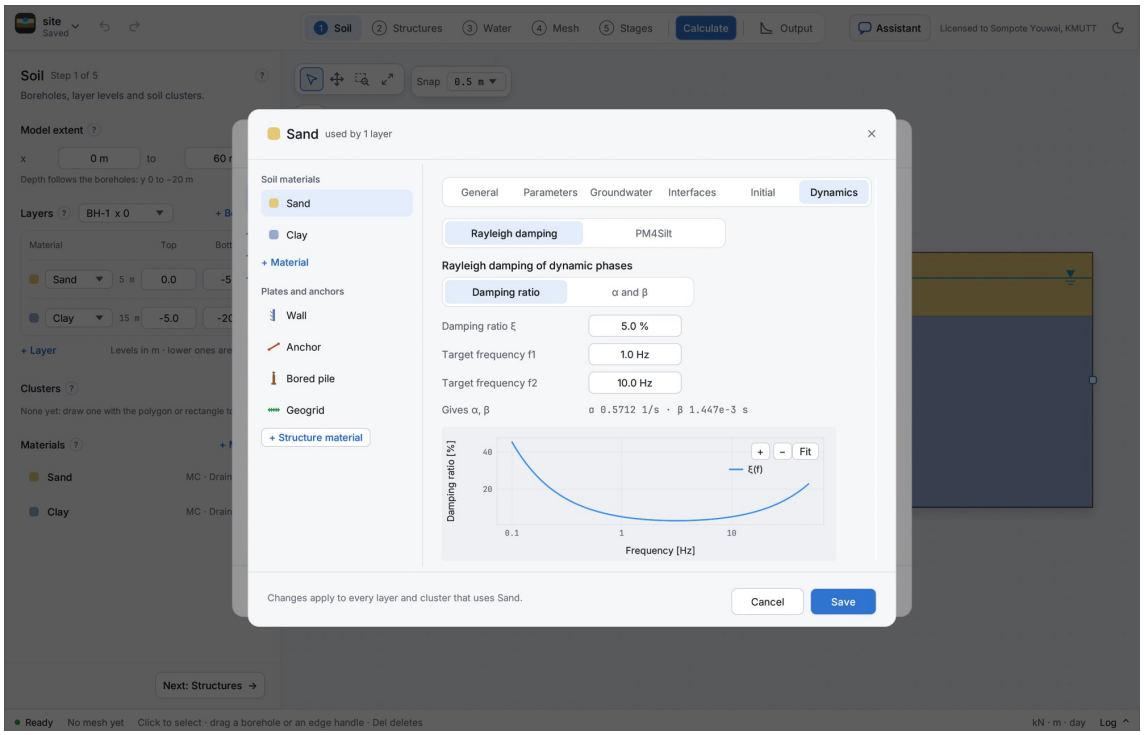


Figure 20. Rayleigh damping of a soil: the target ratio and frequencies, and the damping curve.

8.5 The dynamic phase

1. Press **+ Phase**, name it “Earthquake” and set its **Type** to Dyn. Set the **Duration** to 8 s.
2. Under **Time integration** enter a time step of 0.01 s, or leave the switch on automatic to take it from the signals. The scheme **Average** is the Newmark average acceleration method without numerical damping, and **Damped** adds a little damping of the highest frequencies.
3. Under **Boundaries** choose Free-field for x min and x max, which lets the sides move as a free soil column and absorbs the waves that reach them, and Compliant for the base, which takes the earthquake as an outcrop motion and lets downward waves leave. A rigid base (None) takes the record as the motion within the ground instead.
4. Choose the signal for the horizontal base motion.
5. Pick the points to record with the monitor tool: A under the footing, B on the free surface, C at the base of the sand and D at the base of the model.

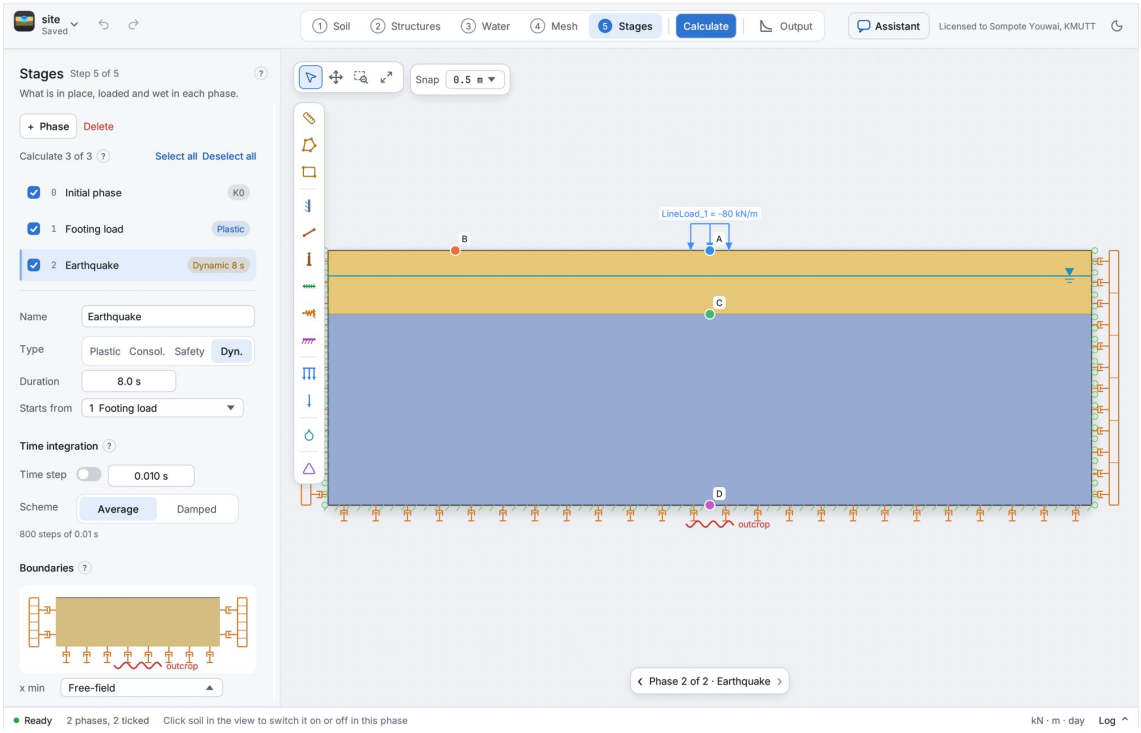


Figure 21. The dynamic phase: duration, time step, scheme and boundaries. The picture shows the free-field columns at the sides, the compliant base with its outcrop motion, and the monitored points A to D.

Boundary	Use it for
Viscous (sides and base)	A source inside the model, such as a machine or pile driving: the dampers absorb outgoing waves
Free-field (sides)	Earthquakes: the sides follow a free soil column and absorb what the structure radiates
Tied (sides)	Earthquakes on level, layered ground: the two sides move together, as in a laminar box
Compliant base	An earthquake given as the motion of a rock outcrop: the base lets downward waves through
None (base)	An earthquake recorded within the ground at the base level, applied as a rigid base motion

8.6 Results

The run takes 800 steps of 0.01 s and about 19 s on a laptop. In Output choose the **Dynamics** tab (Figure 22), which shows the peak acceleration or velocity in the model, the peak of the base input and **Time histories**, a window with the acceleration, velocity or displacement at the monitored points and, with **Spectrum (5 %)**, their response spectra. **Export CSV...** writes the histories to a file.

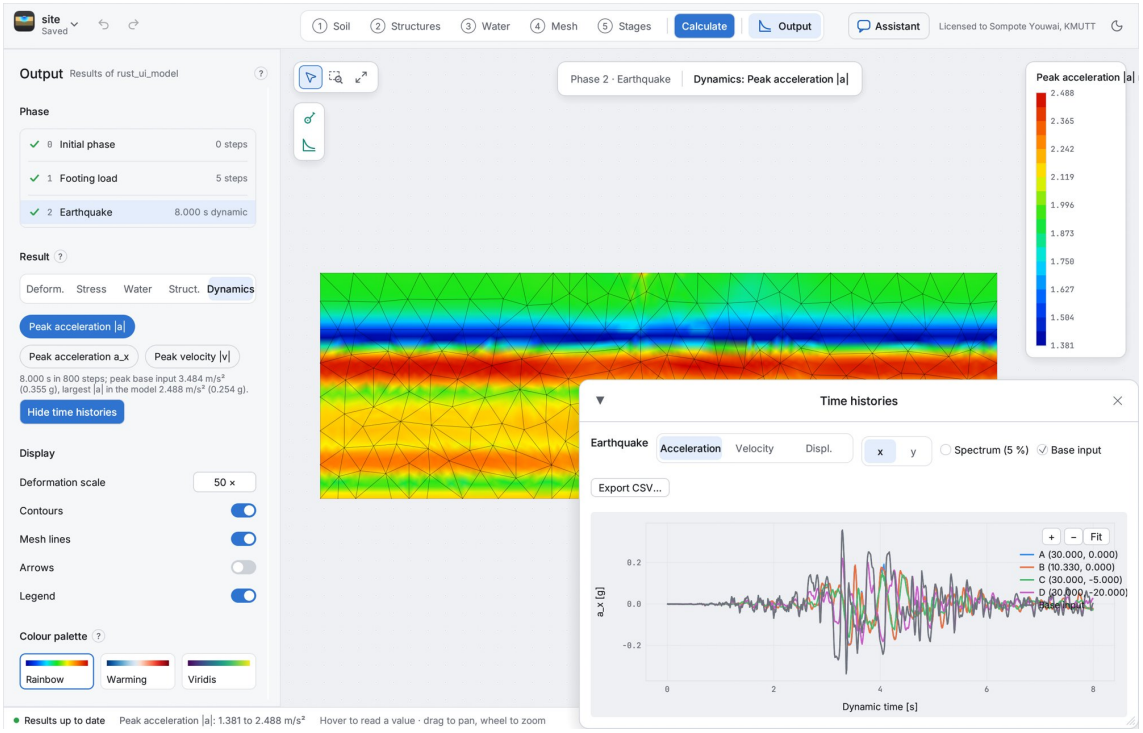


Figure 22. Output of the dynamic phase: peak acceleration in the model and the time histories at the monitored points.

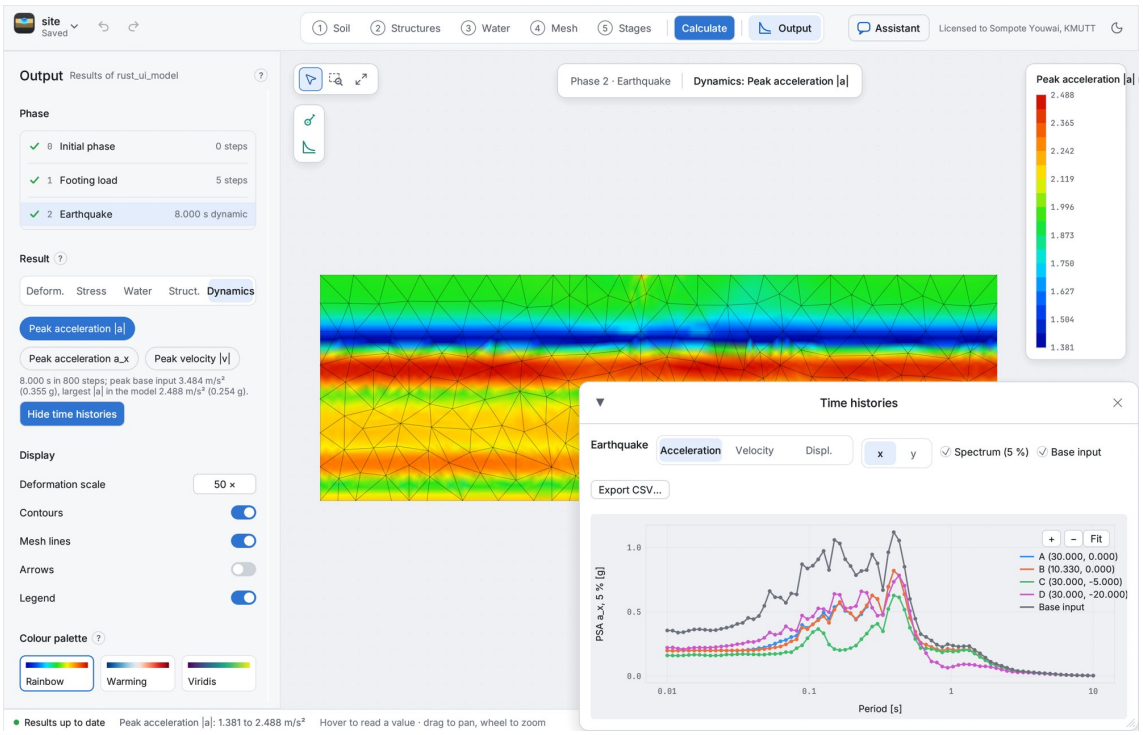


Figure 23. Response spectra (5 % damping) at the monitored points.

Point	Peak horizontal acceleration	Peak horizontal displacement
Base input (outcrop motion)	3.48 m/s² (0.355 g)	
D base of the model, y = -20 m	2.17 m/s² (0.221 g)	47 mm
C base of the sand, y = -5 m	1.57 m/s² (0.160 g)	54 mm
B free surface, x = 10 m	1.94 m/s² (0.198 g)	55 mm

Point	Peak horizontal acceleration	Peak horizontal displacement
A under the footing	1.94 m/s² (0.197 g)	55 mm

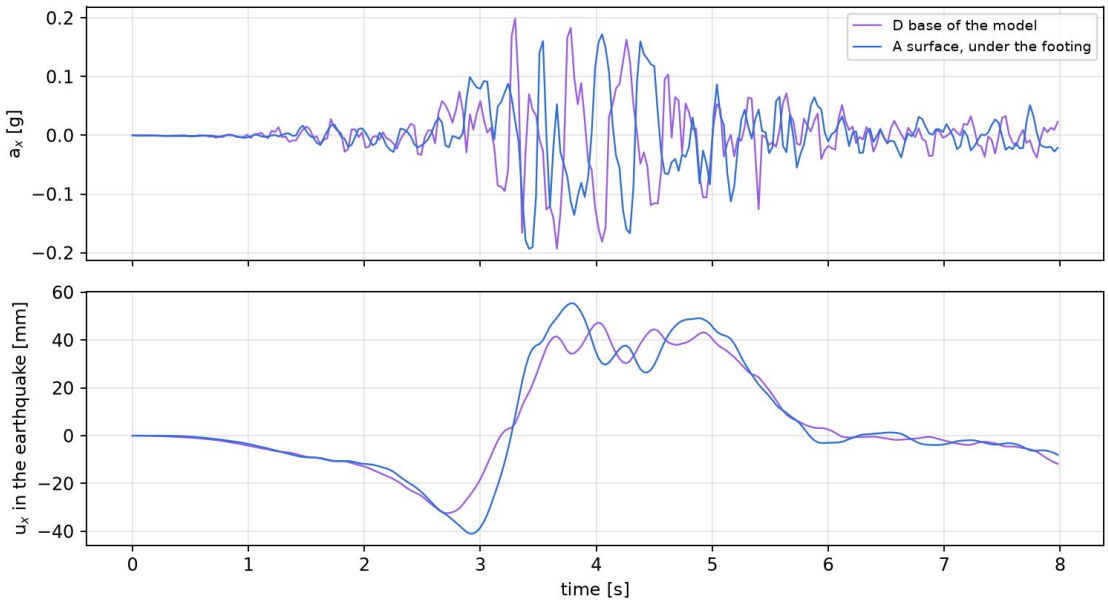


Figure 24. Horizontal acceleration and displacement at the base of the model and under the footing.

The peak acceleration at the surface is 0.198 g against 0.355 g in the outcrop record, so this soft deposit passes on about 56 % of the peak. The footing and the free surface give nearly the same horizontal peak, which shows that a light strip footing does not change the horizontal motion, while the footing adds a vertical acceleration of 0.034 g by rocking. No stress point reached the Mohr-Coulomb strength in this run, so the response is that of a damped elastic deposit, and a stronger motion or a weaker clay would bring plastic points and permanent movement.

The stiffness values of this example are round numbers chosen for a short run, not measured small-strain values, and a constant stiffness with Rayleigh damping does not reduce the stiffness as the strain grows. For a design, take the stiffness from shear wave velocities, check the response against a one-dimensional site response analysis, and use several records.

8.7 The same case through MCP

The dynamic settings have their own tools and options. The script `manual/cases/dynamic_site/make_case.py` builds and runs this case, and the calls that differ from a static model are these:

```
m.tool("update_soil_material", {"material": 0, ..., "damping": {"xi": 0.05, "f1": 1.0, "f2": 10.0}})
m.tool("add_signal", {"name": "Gilroy 1989", "kind": "table",
                      "file": "gilroy_1989.txt", "drift_correction": True})
m.tool("set_monitors", {"points": [[30, 0], [10, 0], [30, -5], [30, -20]]})
m.tool("add_phase", {"name": "Earthquake", "calc": "dynamic",
                     "dynamic": {"time": 8, "dt": 0.01, "x_min": "free-field", "x_max": "free-field",
                                 "y_min": "compliant", "base_x": 0, "base_scale": 9.81}})
m.tool("calculate")
a = m.tool("get_dynamic", {"phase": 2, "quantity": "a"})      # histories and peaks; also "v", "u"
```

A request to an assistant can be as short as: “In Tellus, put a 3 m strip footing with 80 kPa on 5 m of sand over 15 m of clay, 60 m wide, water at 2 m. Shake it with the record gilroy_1989.txt (in g) as an outcrop motion for 8 s with free-field sides and 5 % damping at 1 and 10 Hz. Give me the peak accelerations at the surface and the base, and the time history under the footing.”

9 Published case studies and calibration

9.1 Why the program is calibrated, and on what

A finite element program is only as good as the evidence that its answers are right, so Tellus 2D is checked on three levels. Verification tests compare it with closed-form solutions, which shows that the equations are coded correctly. Lessons of the PLAXIS 2D Tutorial Manual compare it with an established program on fully specified input, which removes the uncertainty of field data. Field case histories compare it with measurements, which shows what accuracy to expect in practice, where the soil parameters are never fully known. This chapter summarises the cases, taken from the calibration paper of the program, its dynamic paper and the embedded pile report, so you can find a case close to your own problem, see how it was modelled, and judge how far to trust a similar analysis. The scripts of all cases are in the source under `soil_FEM/paper` and run through `tellus --mcp` like the case studies of Chapters 7 and 8.

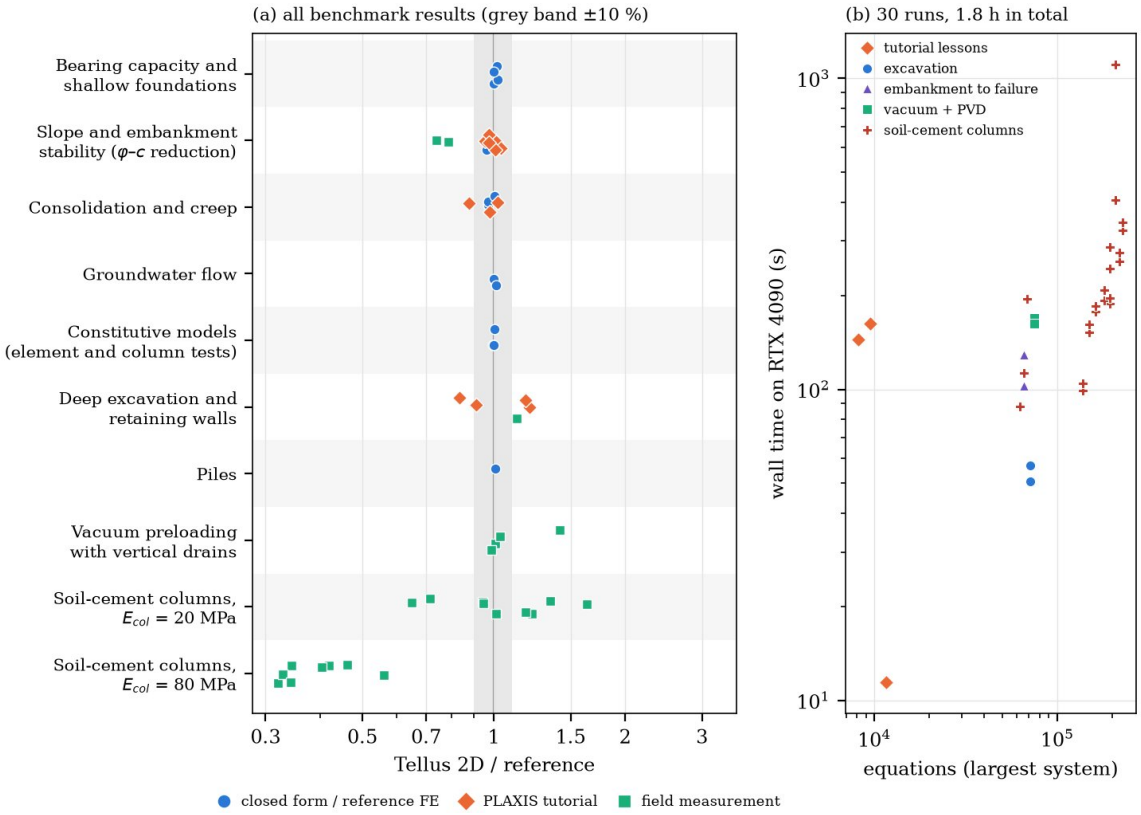


Figure 25. Summary of the static calibration: (a) every benchmark result as the ratio of Tellus 2D to its reference, with a band of ±10 %, and (b) the wall time of each run on a cloud GPU against the size of the system of equations.

Application	Calculation and models	Benchmarks	Agreement
Bearing capacity	Plastic, MC and HS, undrained B and C, phi-c reduction	Footing against a reference FE code, strip collapse (2 + π) su, strip factor of safety	Within 2 %
Slope and embankment stability	phi-c reduction, drained and undrained A, HS and Soft Soil	Griffiths and Lane slope, PLAXIS Lessons 2 and 9, Muar to failure	Factor of safety within 4.3 %. Muar fails 1.2 to 1.4 m early
Consolidation and creep	Biot consolidation, Soft Soil and Soft Soil Creep	Terzaghi, creep settlement, PLAXIS Lesson 9	Within 3 %. Lesson 9 peaks -2 and -12 %, dissipation 3 times faster

Application	Calculation and models	Benchmarks	Agreement
Groundwater flow	Steady confined and unconfined flow	Discharge of an aquifer and of a dam	Within 1.2 %
Excavation and walls	Staged excavation, HS, plates, struts, interfaces, pore pressure profiles	PLAXIS Lesson 3, Sukhumvit 21 m excavation	9 to 17 % dry, 18 to 21 % submerged. Sukhumvit +13 %
Embedded piles	Pile row with skin, base and lateral springs	Closed-form stiffness and capacity, PLAXIS piled abutment	Element within 1 %. Abutment: same shape, lower forces at an earlier stage
Vacuum preloading	Consolidation with vacuum, equivalent drain permeability, Soft Soil	Suvarnabhumi TV2	Within 3 % at three depths
Soil-cement columns	Column walls, undrained lifts and consolidation	Lian-Yun-Gang, nine measured groups	0.65 to 1.64 of the group means with E col = 20 MPa
Dynamics	Newmark, Rayleigh damping, free-field and compliant boundaries, PM4Silt	Wave solutions, DEEPSOIL, Strata, SHAKE91, Garner Valley, Austrian Dam, Bangkok basin	Exact solutions within 3 %. Field cases discussed in Section 9.7

9.2 Verification against closed-form solutions

The solver has 82 verification tests, which all pass on the calibrated version. Table values below are a selection. They show the size of the numerical error of the program itself, which is a few percent at most and much smaller than the uncertainty of soil parameters.

Test	Tellus 2D	Reference
Footing on sand over clay, 50 kPa, largest displacement	9.12 mm	9.125 mm (reference FE code)
Terzaghi consolidation, degree at Tv = 0.2, one-way drainage	0.490	0.505 (analytical)
Strip on undrained clay, collapse load, Undrained B / C	104.8 / 104.3 kPa	(2 + π) su = 102.8 kPa
Strip at half its collapse load, factor of safety	2.041	2.039
Griffiths and Lane 2:1 slope, factor of safety	1.322	1.376 (Bishop), 1.4 (their FE)
Unconfined flow through a rectangular dam, discharge	3.037 m^3/day/m	3.0 (Dupuit-Charny)
Hardening Soil, 10 m column under 100 kPa	30.46 mm	30.39 mm
Hardening Soil, triaxial failure, compression / extension	-372.86 / -26.06 kPa	-372.86 / -26.06 kPa (reference FE)
Soft Soil Creep, 10 m layer, settlement from 100 to 1,000 days	0.0920 m	0.0917 m
Embedded pile row, axial head settlement in a rigid soil	0.2341 mm	0.2341 mm (bar on springs)
Embedded pile row, lateral head deflection	2.407 mm	2.392 mm (Hetényi)
Embedded pile row, axial capacity in compression	504.1 kN/m	500.0 kN/m
Updated mesh, elastic column compressed by 30 %	0.5906 m	0.5931 m (logarithmic strain)

The slope of Griffiths and Lane comes out 4 % below Bishop's value, on the safe side, and a factor of safety from a phi-c reduction changes by up to a tenth with the mesh. Keep this in mind when a calculated factor is close to the required one.

9.3 Lessons of the PLAXIS 2D Tutorial Manual

Three plane-strain lessons of the PLAXIS 2D Tutorial Manual were repeated with the manual's geometry, materials, phases and mesh density, and their results were read off the manual's figures to about 2 to 3 % of each axis (Figure 26). They are good templates for your own models, because each is a complete, fully specified problem.

Lesson	What it is	How it is modelled in Tellus	Result against PLAXIS
2 Embankment stability	4 m sand embankment on 6 m of clay, water at the surface, drained and undrained clay	Hardening Soil, phi-c reduction after construction. Two models, because a material cannot change between phases	Factor of safety 1.833 against 1.81 (drained) and 1.330 against 1.39 (undrained A)
3 Submerged excavation	20 m deep, 30 m wide pit inside 30 m diaphragm walls with one strut, stages 2 and 3 dug under water	Hardening Soil, undrained clay over sand, plate with interfaces (R inter 0.5 and 0.67), a spring as the strut, water levels in the pit	Wall displacement within 9 to 17 % in the dry stages, 21 and 18 % larger in the two submerged stages
9 Road embankment	4 m embankment in two lifts on 3 m of peat and 3 m of clay over sand, consolidation and safety after each lift	Soft Soil, undrained A, consolidation phases of 2, 30, 1 and 22.5 days, a Safety phase branching from each	Pore pressure peaks 25.1 and 27.9 against 25.8 and 31.7 kPa, settlement 0.217 against 0.213 m, factors of safety within 4 %

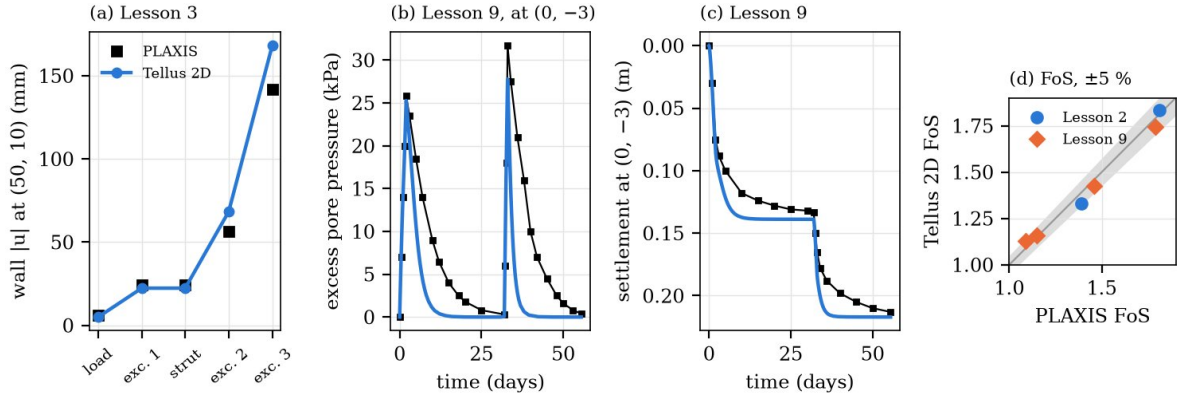


Figure 26. PLAXIS tutorial lessons: (a) Lesson 3, wall displacement at the end of each phase, (b) and (c) Lesson 9, excess pore pressure and settlement under the embankment centre against time, (d) all factors of safety of Lessons 2 and 9. Black squares are PLAXIS.

Two differences are worth knowing. In Lesson 3 the wall moves about a fifth more than in PLAXIS in the submerged stages, and the difference disappears when the clay interfaces are given the full soil strength, so it lies in the response of weak interfaces against undrained Hardening Soil clay. In Lesson 9 the excess pore pressures dissipate about three times faster than in PLAXIS, because Tellus keeps the permeability constant, whereas PLAXIS lowers it with the void ratio and above the water table. Final settlements agree, and the time to reach them is shorter in Tellus, so treat consolidation times in very compressible soil with care.

9.4 Field case histories

Four published case histories were modelled with the parameters of their sources and no tuning to the measurements. Table rows give the essentials, and Figures 27 to 30 show the comparisons.

Case	Model	Tellus 2D	Measured
Sukhumvit station, Bangkok MRT: 21 m top-down excavation in a 1 m diaphragm wall (Likitlersuang et al., 2013)	Half model, Hardening Soil for seven layers, undrained A clays, wall with undrained interfaces, slabs cast in stages, deep-well drawdown profile. 4,392 elements	Wall deflection 40.7 mm at -17.9 m	36 mm at 19 to 20 m (inclinometer IN4)
Muar trial embankment built to failure on soft marine clay (Indraratna et al., 1992)	Half model, Soft Soil for four clay layers, fill in 0.5 m lifts placed undrained and consolidated 8.75 days each. 2,889 elements	Fails at 4.07 and 4.33 m with two parameter sets	Fails at about 5.5 m
Suvarnabhumi TV2: vacuum preloading with vertical drains on soft Bangkok clay (Bergado et al., 1998)	Half model, Soft Soil, under-hydrostatic pore pressure profile, equivalent vertical permeability of the drains, measured membrane vacuum. 3,131 elements	Settlement 0.979, 0.691 and 0.361 m at the surface, 3 m and 6 m at day 145	0.97, 0.70 and 0.35 m at day 144
Lian-Yun-Gang expressway: embankments on soil-cement columns (Chai et al., 2002)	Half model, column grid as plane-strain walls of the same area ratio, Soft Soil clay, lifts and a final long consolidation. 5,769 to 9,493 elements	0.11 to 0.34 m with E col = 20 MPa, 0.04 to 0.18 m with 80 MPa	0.03 to 0.54 m at many sections

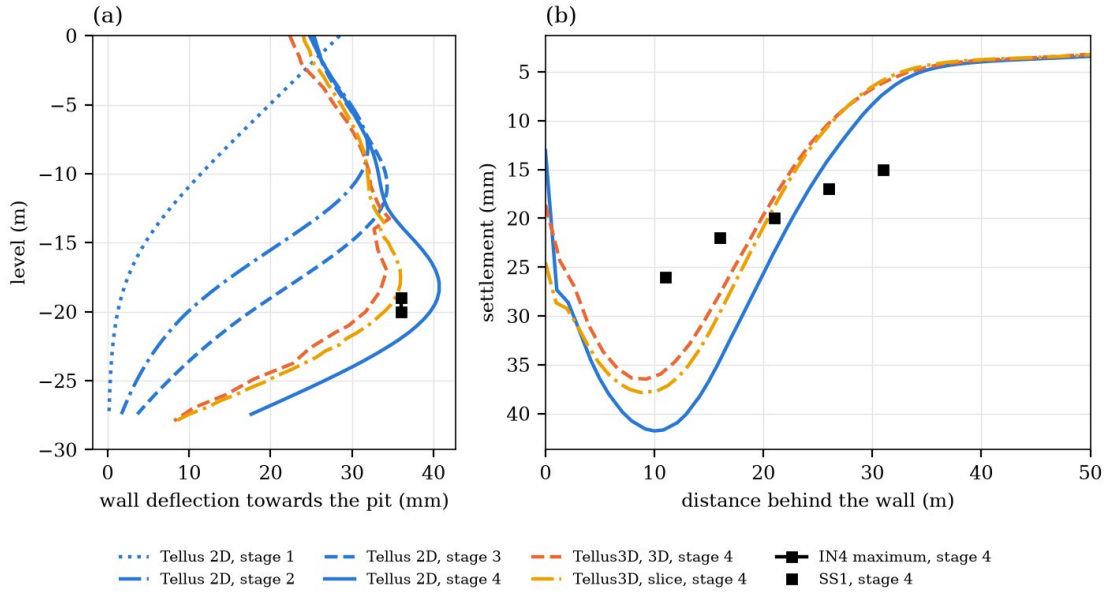


Figure 27. Sukhumvit station excavation: (a) wall deflection after each stage against the measured maximum, and (b) surface settlement behind the wall against the settlement array.

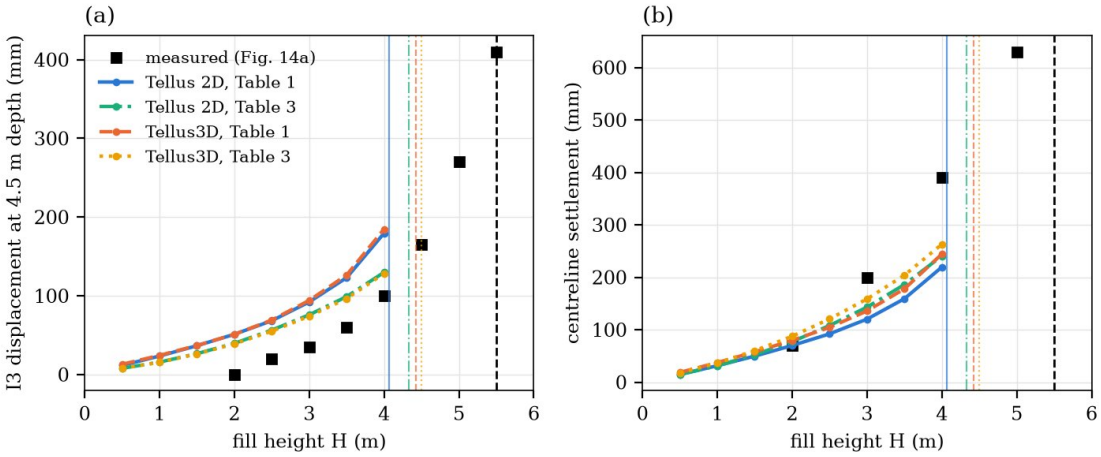


Figure 28. Muar embankment raised to failure: (a) lateral displacement at 4.5 m depth and (b) centreline settlement against fill height. Vertical lines mark the failure heights.

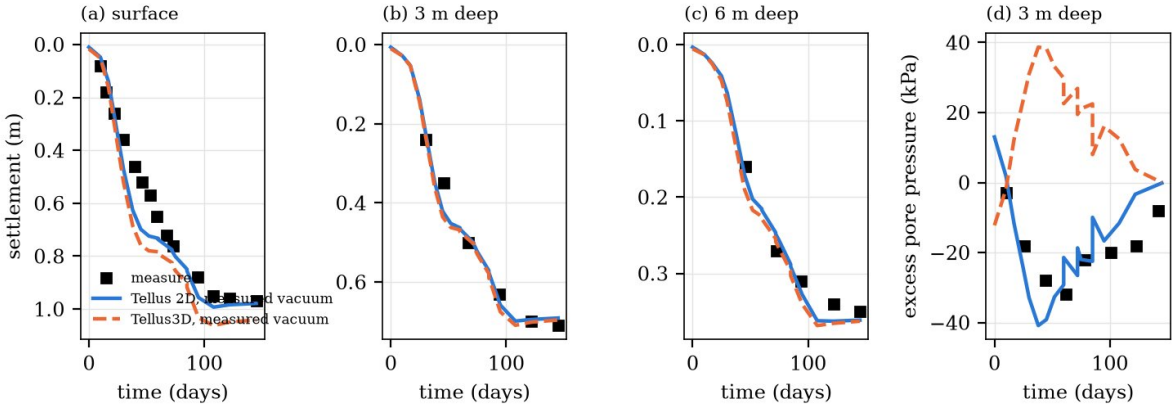


Figure 29. Vacuum preloading of TV2 under the centre of the treated area: settlement at the surface, 3 m and 6 m, and excess pore pressure at 3 m.

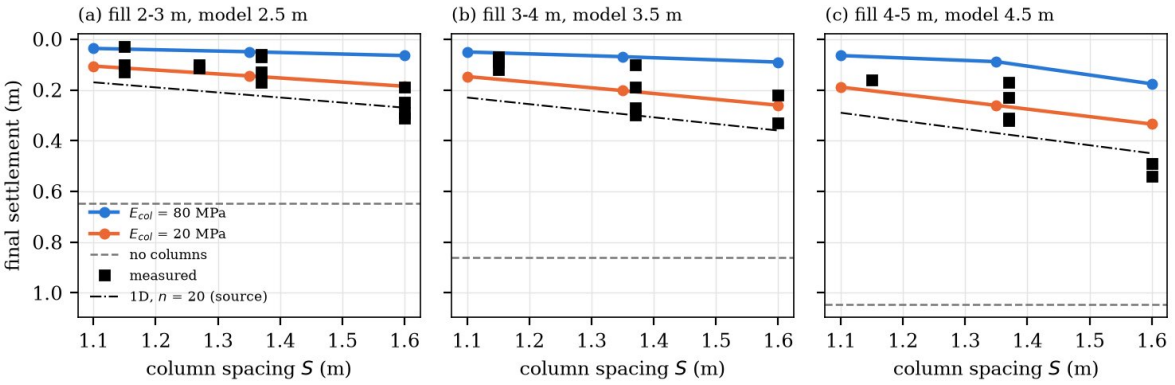


Figure 30. Lian-Yun-Gang soil-cement columns: final settlement against column spacing for three ranges of fill height, with column moduli of 80 and 20 MPa and without columns.

9.5 What the field cases teach

The excavation shows the value of the right pore pressures. With the measured drawdown the wall deflects 40.7 mm against 36 mm measured, and with hydrostatic water it deflects 87.8 mm, more than twice as much, so in Bangkok subsoil the pore pressure profile matters more than any other input. The model also over-predicts the deflection of the first stage, 28.5 mm against 14 to 18 mm, because the temporary prop of that stage is not in it, and its settlement trough is narrower and deeper than measured.

The embankment shows a limit of the soil model. Tellus fails about 1.2 to 1.4 m early, most likely because the Soft Soil model fails on the Mohr-Coulomb hexagon, which is weaker in plane strain than the Cam-clay surface of the original analysis, so a stability result on soft clay from this model is on the safe side. The vacuum trial shows that final settlements can be matched within 3 % when the vacuum history is the measured one, while the early settlement is too fast, and that a vacuum history calibrated for another model does not transfer. The soil-cement columns show how much one assumed parameter can matter: the common rule E = 100 qu makes the model two to four times too stiff, and a modulus ratio of about 45 to the soft clay fits the field.

In every case the largest differences come from inputs that the sources do not fix, not from the numerical method. This is the normal situation in practice, so run your own model with a range of the uncertain parameters and report the range.

9.6 Embedded pile rows under axial and lateral load

The embedded pile row was calibrated on the PLAXIS case study of a piled bridge abutment in an embankment on soft soil (Figure 31). Three rows of 0.54 m piles at 2.4 m centres, two raked forwards and one backwards, carry the abutment and are loaded axially by the soft layers that settle around them and laterally by the soil that moves towards the cut under the bridge deck. The source gives the normal force, shear force and bending moment along each pile from PLAXIS 2D and from PLAXIS 3D. The model uses Hardening Soil for the sands and Soft Soil Creep for the clay and peat, 1,833 elements and consolidation phases for every construction stage, and the piles use the default interface stiffness factors.

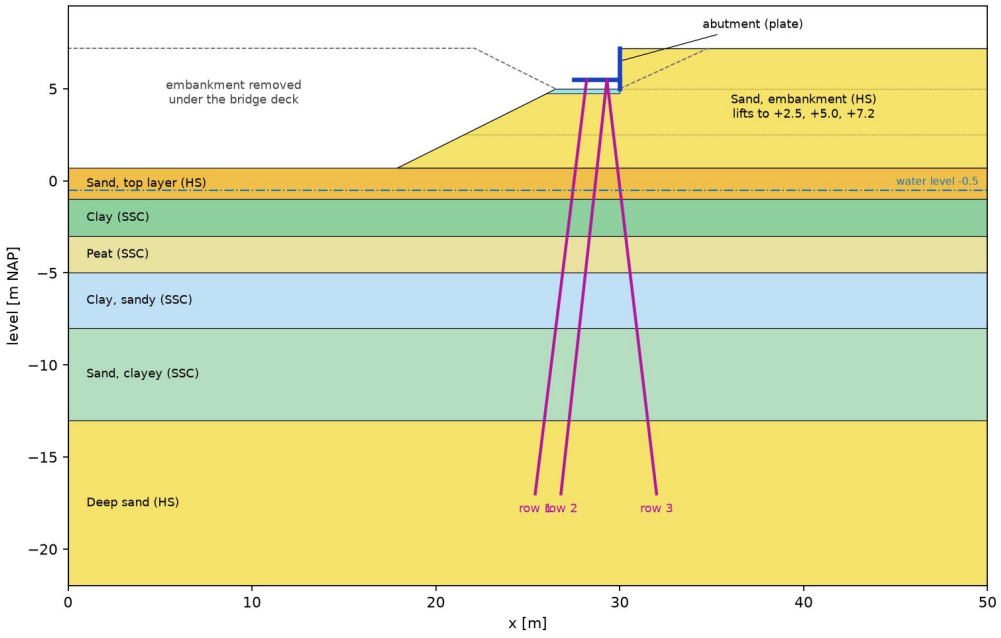


Figure 31. The piled abutment model at its final stage.

Row	Force	Tellus 2D	PLAXIS 2D	PLAXIS 3D
1	Largest N [kN per pile]	-331	-336	-443
	Largest M [kNm per pile]	-236	-307	-313
2	Largest N [kN per pile]	-905	-1188	-1053
	Largest M [kNm per pile]	-230	227	-258
3	Largest N [kN per pile]	309	366	526
	Largest M [kNm per pile]	-290	-640	-593

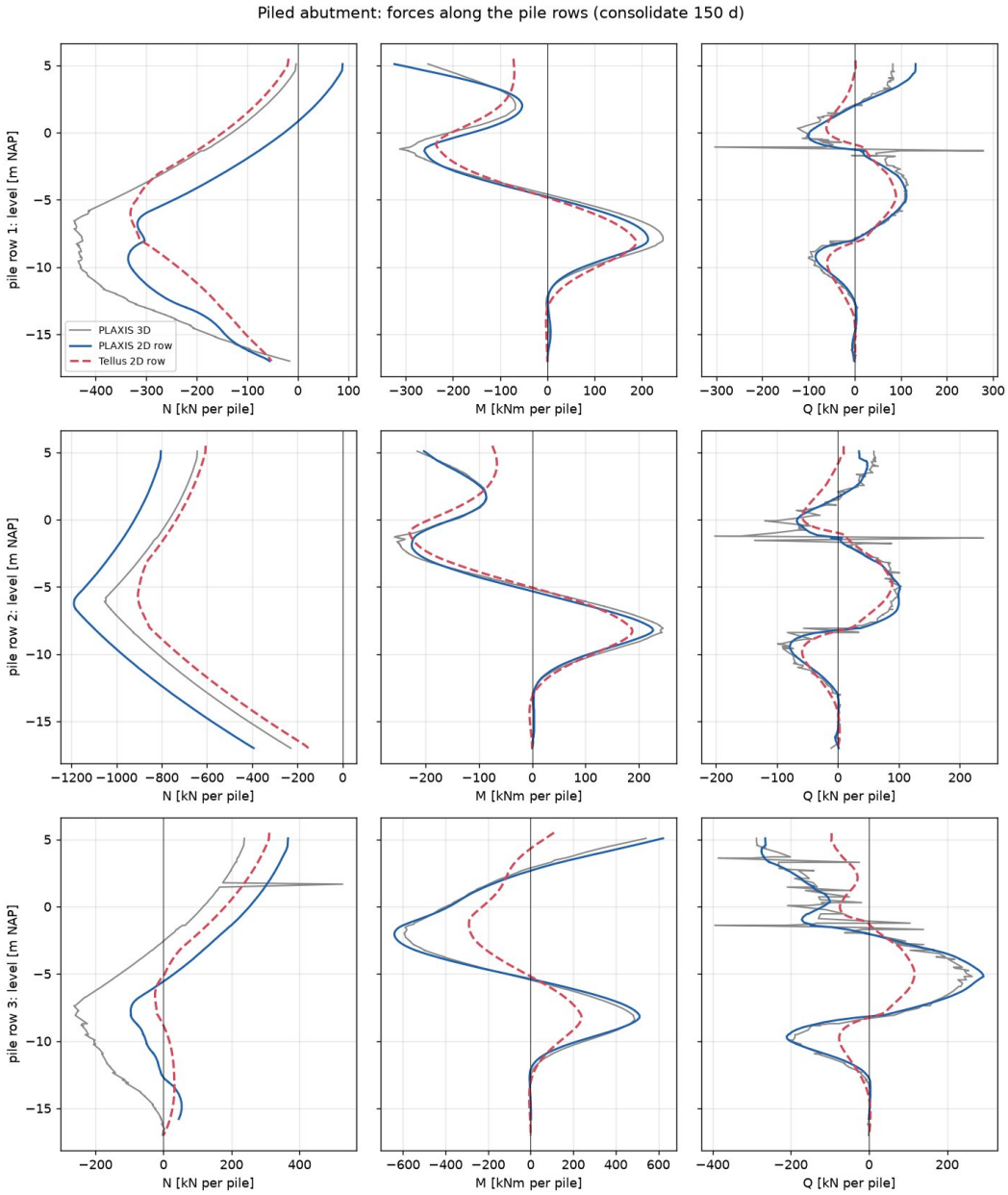


Figure 32. Normal force, bending moment and shear force along the three pile rows: Tellus 2D against PLAXIS 2D and PLAXIS 3D.

Tellus gives forces of the same sign, shape and position as both PLAXIS results, and lower magnitudes. The comparison is not yet like for like. The Tellus result is for 150 days after construction with small displacements and 93 kPa of excess pore pressure left, whereas the source consolidates to below 5 kPa with an updated mesh, and the Tellus pile forces were still rising. The moments at the pile heads are also much smaller than in PLAXIS, which needs a check of the rigid head connection. Use

embedded pile rows with this in mind: the element itself is verified within 1 %, and the default interface factors apply to bored piles with a spacing of 2 to 8 diameters.

9.7 Dynamic analysis

The dynamic part has its own verification and validation. Exact solutions of wave propagation and damping are reproduced within 3 %. Against other programs, the surface motion of DEEPSOIL is matched within 1.5 %, the spectra of Strata for ten records with a mean logarithmic ratio of 0.008, and the SHAKE91 sample problem within 2 %. In the PLAXIS tutorial lesson of a building under an earthquake the period agrees within 5 %, although the static deflection with HS small is 27 % larger than in the tutorial.

Case	What was compared	Result
Garner Valley downhole array, 12 records	Surface spectra from the recorded downhole motion, one-dimensional column	Over-predicted by a mean logarithmic ratio of 0.33 at 2 % damping, 0.23 with the damping matched at the site frequency
PM4Silt element tests	Critical-state strength and cyclic strength against the FLAC implementation	Strength exact, number of cycles within a factor of about 1.5
Austrian Dam, 1989 Loma Prieta earthquake	Excess pore pressure and crest acceleration with PM4Silt	Pore pressures of the measured magnitude, crest 0.58 to 0.72 g against 0.7 g in FLAC. Crest settlement could not be compared
Bangkok basin, 54 km section, 2025 Mw 7.7 Myanmar earthquake	Recorded velocity and amplification	About 10 cm/s and amplification near 1 s reproduced, within 10 % of one-dimensional columns

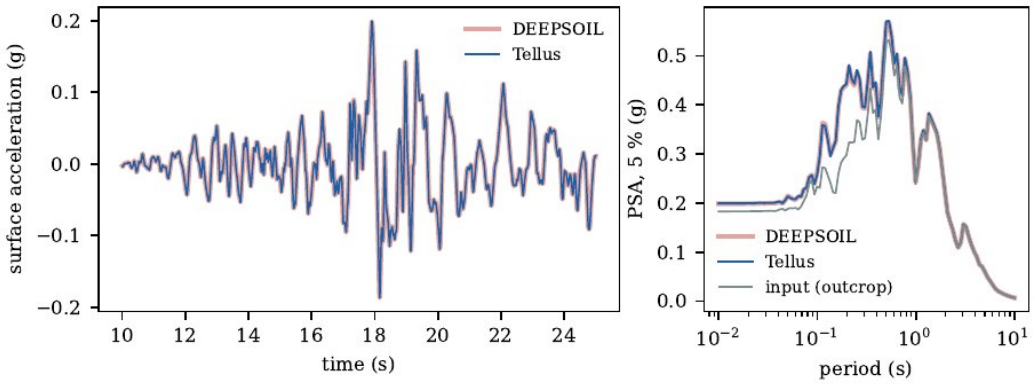


Figure 33. Site response against DEEPSOIL for the same layered column and input motion.

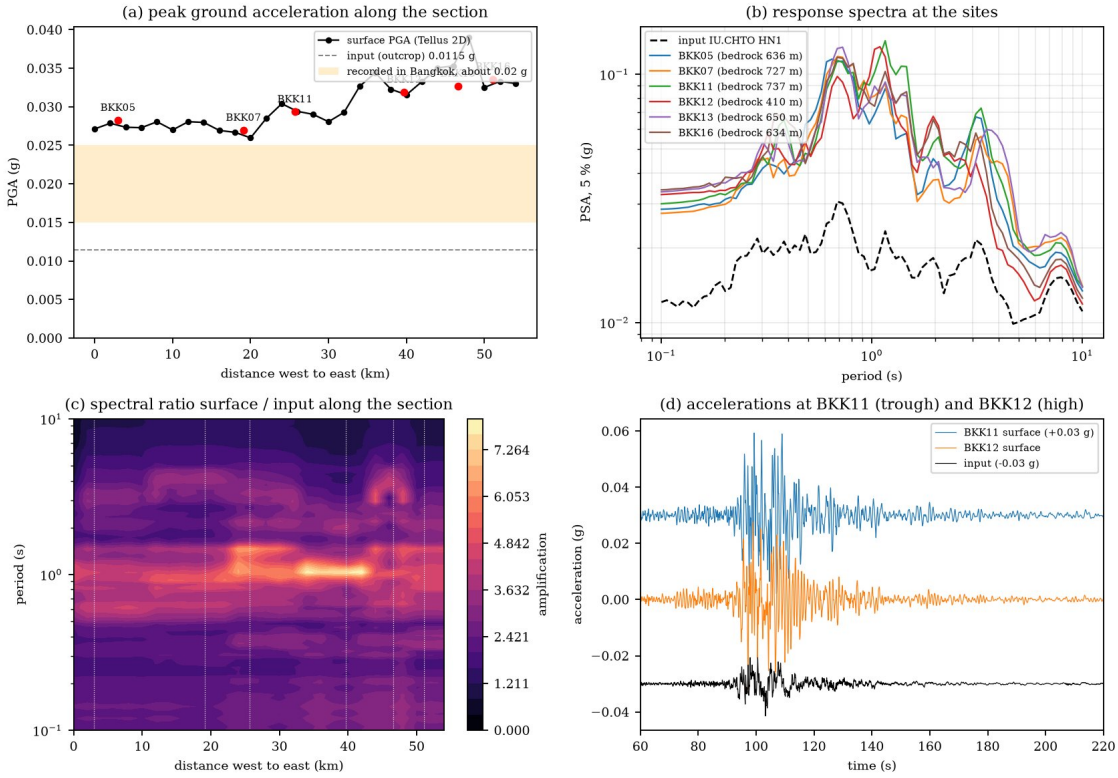


Figure 34. The Bangkok basin section in the 2025 Myanmar earthquake.

The dynamic validation has clear limits. Linear soil with Rayleigh damping over-predicts the surface motion at Garner Valley unless the damping is matched at the site frequency, and at Austrian Dam the loose unsaturated fill began to flow in strong shaking, which an implicit solution cannot follow, so the crest settlement of that case is not validated. Use the dynamic phases for accelerations, spectra and pore pressure build-up, and be careful with permanent displacements of soil that is close to failure.

9.8 Using a published case as a template

1. Find the case closest to your problem in the tables of this chapter and open its script in `soil_FEM/paper/cases`, `soil_FEM/paper/new_cases`, `soil_FEM/paper/plaxis_tutorial` or `soil_FEM/paper/dynamic`.
2. Run it once unchanged with `python3 run_case.py` and open the .tellus file it saves in the window, to see the layers, structures and phases.
3. Copy the script, replace the geometry and the soil parameters with yours, and keep the modelling choices that the case needed, such as the pore pressure profile of an excavation or the lifts and consolidation times of an embankment.
4. Or give the script to Claude Code and ask it to adapt the case, for example “use this Sukhumvit model for a 15 m excavation with a 0.8 m wall and these layers”.

References of this chapter: Likitlersuang et al. (2013), Soils and Foundations 53(5), 756-773. Indraratna, Balasubramaniam and Balachandran (1992), Journal of Geotechnical Engineering 118(1), 12-33. Bergado, Chai, Miura and Balasubramaniam (1998), Geotechnical Engineering Journal (SEAGS) 29(1). Chai, Liu and Du (2002), Field properties and settlement calculation of soil-cement column improved soft subsoil, a case study. PLAXIS 2D Tutorial Manual. PLAXIS (2012), Embedded Pile Row in Plaxis 2D, case study. The full comparisons are in the calibration paper (soil_FEM/paper/paper.pdf), the dynamic paper (soil_FEM/paper/dynamic/dynamic.pdf) and the embedded pile report (paper/embeddedpile.pdf).

10 Structures and their parameters

10.1 Structures in a plane-strain model

A Tellus 2D model is a slice one metre thick, so every structure is entered per metre run of the section. A continuous wall or slab has its real stiffness per metre. A row of separate members, such as struts, anchors or piles at some spacing, is smeared over that spacing, so the stiffness and the strength of one member are divided by the spacing, and a force that the program reports is multiplied by the spacing to get the force in one member. Table rows below list the structural elements, and the sections that follow explain the parameters of the three that need most care: plates, embedded pile rows and geogrids.

Element	What it models	Joined to the soil by	Carries
Plate	Walls, sheet piles, slabs, tunnel linings, footings	Its own nodes on a mesh line, with optional interfaces on each side	Axial force, shear force, bending moment
Embedded pile row	A row of piles, with soil free to move between them	Skin, lateral and base springs, without splitting the mesh	Axial force, shear force, bending moment
Geogrid	Geotextiles and geogrids, soil reinforcement	The soil's own nodes	Axial tension only
Node-to-node anchor	Tie rods, the free length of ground anchors	Its two end nodes only	Axial force, optional strength
Spring support	Struts, fixed-end anchors	One node and a fixed end	Axial force along its direction

All structures are drawn in step **2 Structures** and switched on per phase in step **5 Stages**. A structure carries only the movement that happens after it is switched on, so a strut installed after an excavation stage is not loaded by that stage. Structure materials are edited in the material dialog under **Plates and anchors**, and one material can serve several structures.

10.2 Plates

A plate is a beam along a line of the mesh with three degrees of freedom per node, two displacements and a rotation, and it includes shear deformation (Mindlin beam). Use it for anything that resists bending: a diaphragm wall, a sheet pile, a base slab, a raft or a tunnel lining. Draw it with the plate tool by clicking its two ends, and split a long wall at the points where another structure joins it, so a node lies there.

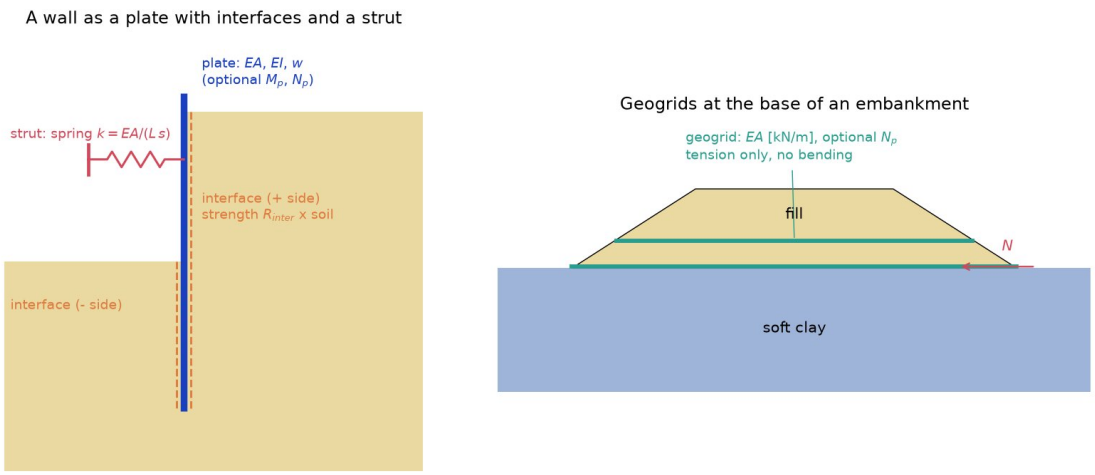


Figure 35. Left: a wall as a plate with an interface on each side and a strut as a spring. Right: geogrids at the base of an embankment.

Parameter	Unit	Meaning	How to get it
EA	kN/m	Axial stiffness per metre run	E times the cross-section area per metre. For a solid wall of thickness d, EA = E d
EI	kNm²/m	Bending stiffness per metre run	E times the moment of inertia per metre. For a solid wall, EI = E d³ / 12
w	kN/m/m	Weight per metre of plate and per metre run	For a wall that replaces soil, (γ wall - γ soil) d, because the soil's weight is already in the mesh. For a slab above ground, γ d
Elastoplastic, Mp	kNm/m	Plastic bending moment: a hinge forms where |M| reaches it	Design moment capacity of the section per metre. Leave the plate elastic unless you want to see hinges form
Elastoplastic, Np	kN/m	Plastic axial force	Axial capacity of the section per metre
Interfaces		A zero-thickness layer between the plate and the soil on each side, positive and negative	Tick both sides for a wall in the ground, none for a slab that should move with the soil
Rayleigh α, β	1/s, s	Damping of the plate in dynamic phases	Usually zero, or a small structural damping. The mass is w / g

The program takes the shear stiffness of a plate from its axial stiffness, kGA = 5/12 EA, so EA and EI should belong to the same section, and the equivalent thickness d = √(12 EI / EA) is a quick check that they do. A plate has no width in the model: it takes no room in the soil and carries no end bearing, which matters for a wall or pile that stands on its toe. The results are the axial force N, the shear force Q and the bending moment M along the plate, per metre run.

10.3 Interfaces along plates

An interface lets the soil slide along a structure and open a gap behind it. Its strength is that of the adjacent soil reduced by the factor **R inter** of the soil material (tab Interfaces): c i = R inter c' and tan φ i = R inter tan φ'. A value of 1 means a fully rough contact, and smaller values a smoother one. The interface stiffness follows from the soil stiffness and a thin virtual thickness, so it needs no input. In an undrained soil the interface is undrained too, and an interface next to excavated soil stops acting.

Contact	Typical R inter
Sand against steel	0.6 to 0.7
Clay against steel	about 0.5
Sand against concrete	0.8 to 1.0
Clay against concrete	0.7 to 1.0
Soil against a geotextile or geogrid (when an interface is modelled)	0.5 to 0.9, from pull-out or shear box tests

These ranges are common starting values, not properties of your site. R inter changes wall deflections and bending moments noticeably in soft clay, so vary it when it matters, and take it from interface shear tests when you have them.

10.4 Worked plate values

Structure	Input	EA [kN/m]	EI [kNm²/m]	w [kN/m/m]
Diaphragm wall 0.8 m, E = 30 GPa, in soil of 18 kN/m³	d = 0.8 m, γ concrete 24	2.4e+07	1.28e+06	4.8
Steel sheet pile with A = 180 cm²/m and I = 40,000 cm⁴/m (example section), E = 210 GPa	A = 0.018 m²/m, I = 4.0e-4 m⁴/m	3.78e+06	8.4e+04	1.4
Base slab 1.0 m, E = 30 GPa, cast on the ground	d = 1.0 m	3e+07	2.5e+06	24.0

Take A and I of a sheet pile from the manufacturer's table for the section you use, per metre of wall. For cracked reinforced concrete under long-term load, many designers reduce EI, and the reduction is a design decision that the program does not make for you.

10.5 Embedded pile rows

An embedded pile row models a line of piles at a regular spacing out of the plane, such as the piles under a wall, an abutment or a raft (Figure 36). It differs from a plate in an important way: a plate is a continuous wall that the soil cannot pass, whereas the pile row lies over the mesh without splitting it, so the soil moves between the piles. The pile is a beam with its own nodes, and each node is joined to the soil under it by a skin spring along the pile, a lateral spring across it, and at the tip a base spring. Draw it with the pile tool from its head to its tip. A part above the ground is a plain beam without springs.

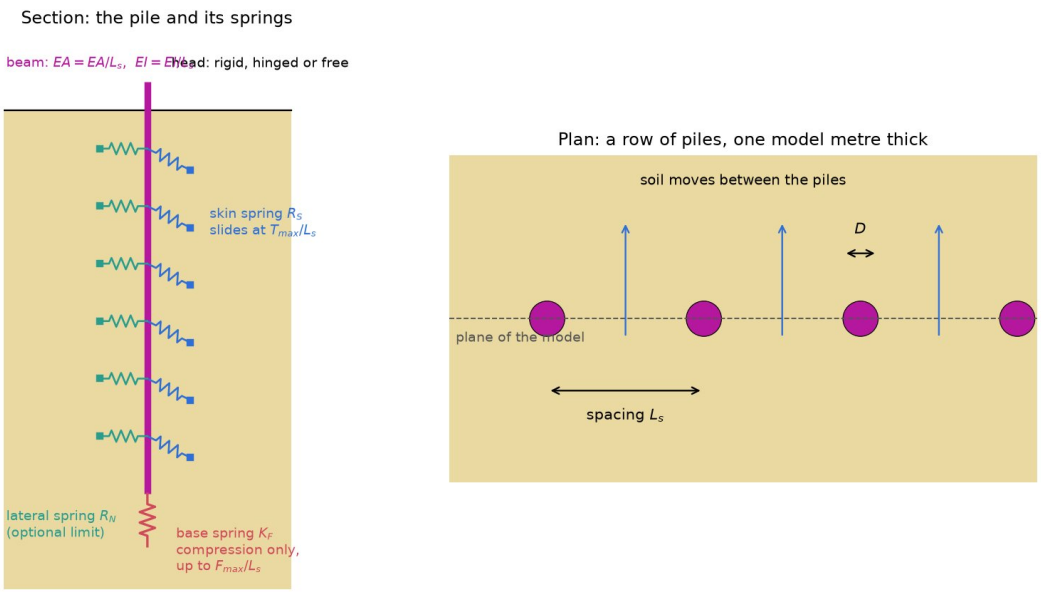


Figure 36. An embedded pile row: the beam and its skin, lateral and base springs in section, and the row in plan.

Parameter	Unit	Meaning	How to choose it
E	kPa	Young's modulus of the pile material	About 30e6 for concrete, 210e6 for steel
Unit weight γ	kN/m³	Weight of the pile	γ pile minus γ soil, because the pile overlaps soil that already has weight. Zero ignores it
Spacing L	m	Centre-to-centre distance of the piles out of the plane	The real spacing. Everything per pile is divided by it
Cross section		Solid circular (D), circular tube (D, wall thickness t), solid square (width) or user-defined (A, I)	The real section. It gives A, I and the equivalent diameter D eq = √(12 I / A) for a user section
Axial skin resistance	kN/m	The largest shaft friction per metre of one pile	Linear: values at the head and the tip. Multi-linear: a table against depth along the pile. Layer dependent: from the soil strength and R inter, up to T max
Lateral resistance	kN/m	The largest lateral soil reaction per metre of one pile	Unlimited (elastic springs) by default. Linear or multi-linear to cap it, for example at an ultimate p-y value
Base resistance F max	kN	The largest tip force of one pile, in compression only	Base capacity q b times the tip area
Interface stiffness		Factors for the axial, lateral and base springs	Default values, or your own three factors after a calibration
Connection (per pile)		How the head is joined	Automatic: rigid to a plate at the head point, otherwise hinged to the soil. Or rigid, hinged, free

10.6 How the pile springs are calculated

The beam has the stiffness of one pile divided by the spacing, EA = E A / L and EI = E I / L. The springs take their stiffness from the shear modulus G of the soil at each node:

```
skin spring      R_S = ISF_RS x G / L          [kN/m per m of pile, per metre run]
```

```
lateral spring   R_N = ISF_RN x G / L
base spring      K_F = ISF_KF x G x R_eq / L   [kN/m per metre run]

default factors  ISF_RS = ISF_RN = 2.5 x (L / D_eq)^-0.75
                 ISF_KF = 25 x (L / D_eq)^-0.75
```

The default factors come from the validation of the embedded pile row by Sluis (2012). They were derived for bored piles under static load, piles that are stiff compared with the soil, and a spacing of 2 to 8 diameters. A closer spacing makes the row act almost as a wall, and a plate is then the better model, while a much wider spacing is outside the range that was checked. With **Layer dependent** skin resistance the limit at each node is T = 2 π R eq (R inter c' + σ'n R inter tan φ') with the current normal stress, so it follows the soil layers and the construction stages, and T max caps it.

Worked example: bored pile D = 0.6 m, E = 30 GPa, spacing L = 2.0 m	Value
EA per metre run = E π D^2 / 4 / L	4.24e+06 kN/m
EI per metre run = E π D^4 / 64 / L	9.54e+04 kNm²/m
L / D	3.33
Default factors: axial, lateral, base	1.013, 1.013, 10.13
Skin resistance for a shaft friction of 50 kPa: T = π D τ	94 kN/m per pile
Base resistance for q b = 2,000 kPa: F max = q b π D^2 / 4	565 kN per pile
Largest axial load of the row, 12 m pile with that friction: (T L pile + F max) / L	848 kN per metre run

10.7 Reading pile results, and the limits of the element

After a calculation the log and the MCP tool `calculate` give for every pile and phase the axial force at the head, the largest bending moment, and what the skin and the base carry, all per metre run. In Output, **Structures** draws N, Q and M along each pile, and `get_plate_forces` with `full: true` lists them with the skin and lateral tractions node by node. Multiply by the spacing for one pile. Soil that settles around a pile drags it down through the skin springs (negative skin friction), which shows as an axial force that grows with depth to the neutral point.

Know three limits. The pile does not displace soil, so installation effects of a driven pile are not modelled and it behaves as a bored pile. The lateral spring is elastic unless you give a limit, so the row is not a tool for the ultimate lateral capacity of a pile, where a p-y analysis or a 3D model is needed. The element is verified against closed-form solutions within 1 %, and its calibration on a piled abutment against PLAXIS gave the same force distributions with lower magnitudes at an earlier stage of consolidation (Section 9.6).

10.8 Geogrids and geotextiles

A geogrid element models any thin reinforcement that works in tension: geogrids, woven geotextiles, steel strips. It lies along a line of the mesh and uses the soil's own nodes, so it moves with the soil, and it carries only axial tension. In compression it goes slack, and it has no bending stiffness. Draw it with the geogrid tool by clicking its two ends, and place it in the phase in which the layer above it is built.

Parameter	Unit	Meaning	How to get it
EA	kN/m	Axial stiffness per metre width, the tension per unit strain	From the manufacturer's tensile test: the tension at a strain divided by that strain, for example T at 2 % divided by 0.02

Parameter	Unit	Meaning	How to get it
Elastoplastic, Np	kN/m	Tensile strength: beyond it the grid stretches plastically and keeps that stretch	The long-term design strength, which is the short-term strength divided by the reduction factors for creep, installation damage and durability

Worked example	Value
Geogrid with a tension of 20 kN/m at 2 % strain	EA = 20 / 0.02 = 1,000 kN/m
Same grid, stiffness for a 50-year load from its isochronous curve, say 14 kN/m at 2 %	EA = 700 kN/m
Short-term strength 80 kN/m, combined reduction factor 2.5	Np = 32 kN/m
Steel strip 50 mm by 4 mm every 0.75 m, E = 210 GPa	EA = 210e6 × 0.0002 / 0.75 = 56,000 kN/m

Polymer reinforcement creeps, so its stiffness depends on the load duration and the temperature. Use the stiffness for the duration that matters, short-term for construction and long-term for the service life, and take the values in the example above only as an illustration of the arithmetic.

10.9 How a geogrid works in the model

A grid carries load only where the soil stretches along it. Under an embankment on soft ground the base spreads sideways, so a grid at the base goes into tension and holds the fill together. Behind a reinforced wall or in a steep slope the grids tie the active wedge to the soil behind it. Just below a footing on stiff ground the soil is compressed sideways, so a shallow grid there stays slack and adds nothing, which is a correct result and not an error. The grid shares its nodes with the soil, so it cannot slide relative to it, and a pull-out failure is not modelled unless the soil around it fails.

When the ground under the grid settles a lot, for example over a void, between piles or columns, or on very soft clay, the grid also carries load by sagging, like a hammock. This membrane action needs the **Updated mesh** option of the phase (Section 12.1), because with small displacements a force along a horizontal grid has no vertical component. The result of a geogrid is its tension N along its length, in kN/m, with the largest value and its position per phase. Compare it with the design strength, and where the tension reaches Np the grid has yielded.

10.10 Anchors and struts in brief

Element	Parameters	Notes
Spring support	Position, angle (degrees anticlockwise from +x, from the node towards the fixed end), k = EA / (L × spacing) in kN/m per metre run	Linear in tension and compression. A strut that goes into tension should be switched off in that phase
Node-to-node anchor	Two end points, EA of one anchor in kN, spacing, and kind: elastic, elastoplastic (strength in tension and compression) or with a residual strength	Carries axial force only and does not touch the soil between its ends, so it suits a tie rod or the free length of a ground anchor

10.11 Structures through MCP

Each structure has its tool, and the material goes with the call. The values below are those of the worked examples of this chapter.

```
m.tool("add_plate", {"start": [15, 0], "end": [15, -18], "interfaces": True,
                     "active_in": "none",
                     "material": {"name": "D-wall 0.8", "ea": 2.4e7, "ei": 1.28e6, "w": 4.8}})

m.tool("add_embedded_pile", {"head": [20, 0], "tip": [20, -12], "connection": "auto",
                             "active_in": "none",
                             "material": {"name": "Bored pile 0.6", "e": 3.0e7, "spacing": 2.0,
                                          "section": 0, "d": 0.6, "gamma": 6.0,
                                          "skin": 0, "t_top": 94.0, "t_bot": 94.0,
                                          "lateral": 0, "f_max": 565.0,
                                          "isf_default": True}})

# ea makes the geogrid material, np makes it elastoplastic
m.tool("add_geogrid", {"start": [5, 0], "end": [35, 0], "ea": 1000.0, "np": 32.0,
                       "active_in": "none"})

m.tool("add_phase", {"name": "Install", "plates": [True], "piles": [True],
                     "geogrids": [True]})
forces = m.tool("get_plate_forces", {"phase": 1, "full": True})   # N, Q, M per metre run
```

In a request to an assistant, give the same data in words and say what is per member and what is per metre, for example: “bored piles 0.6 m in diameter at 2 m centres, 12 m long, shaft friction 50 kPa, base capacity 565 kN per pile, rigidly connected to the raft”.

11 Water and mesh

Step 3 Water and step 4 Mesh take little time to fill in, and both can change a result more than a soil parameter does. The pore pressure decides the effective stress, and with it the strength and the load on every wall. The mesh decides how well the stresses are resolved where they change fast. This chapter explains the options of both steps and shows their effect on the excavation of Chapter 4.

11.1 How Tellus treats water

Tellus splits the pore pressure into two parts. The **steady** pore pressure comes from the groundwater: a water level, a measured profile or a flow calculation. The **excess** pore pressure comes from loading undrained soil and dissipates with time in consolidation phases. The steady part is an input that you define in the Water step and choose per phase in Stages, and the excess part is a result. A change of the steady pore pressure from one phase to the next loads the soil, so lowering the water settles the ground and raising it reduces the effective stresses.

Input	Where	Meaning
Borehole water level	Water, Pore pressure tab	A level line at one elevation. It is the global level, used by every phase that does not choose another
Drawn water levels	Water, + Draw or + Level line	Any number of phreatic lines as polylines. A line continues level beyond its end points
Unit weight of water	Water, Pore pressure tab	γw, 10 kN/m³ by default, or 9.81
Drainage of each side	Water, Flow and drainage tab	Open or closed, for steady flow and for consolidation
Permeability kx, ky	Soil material, Groundwater tab	In m/day, used by steady flow and consolidation
Drainage type	Soil material, General tab	Drained, Undrained (A), (B), (C) or non-porous
Water of a phase	Stages, Water and Pore pressure	Which level the phase uses, and Phreatic or Steady flow

11.2 Water levels

A water level is a phreatic line. Under each point of the line the pore pressure is hydrostatic, p = γw (h - y) with h the level of the line above the point, and it is zero above the line. The borehole level is entered as one number. To draw another level, press **+ Draw** and click its points from left to right, finishing with a double-click or Enter, then edit the points and the name in the panel. **+ Level line** adds a level line across the model that you then move. Draw a level only where the water surface changes, for example down into a dewatered pit and back up outside the wall, as in Figure 37.

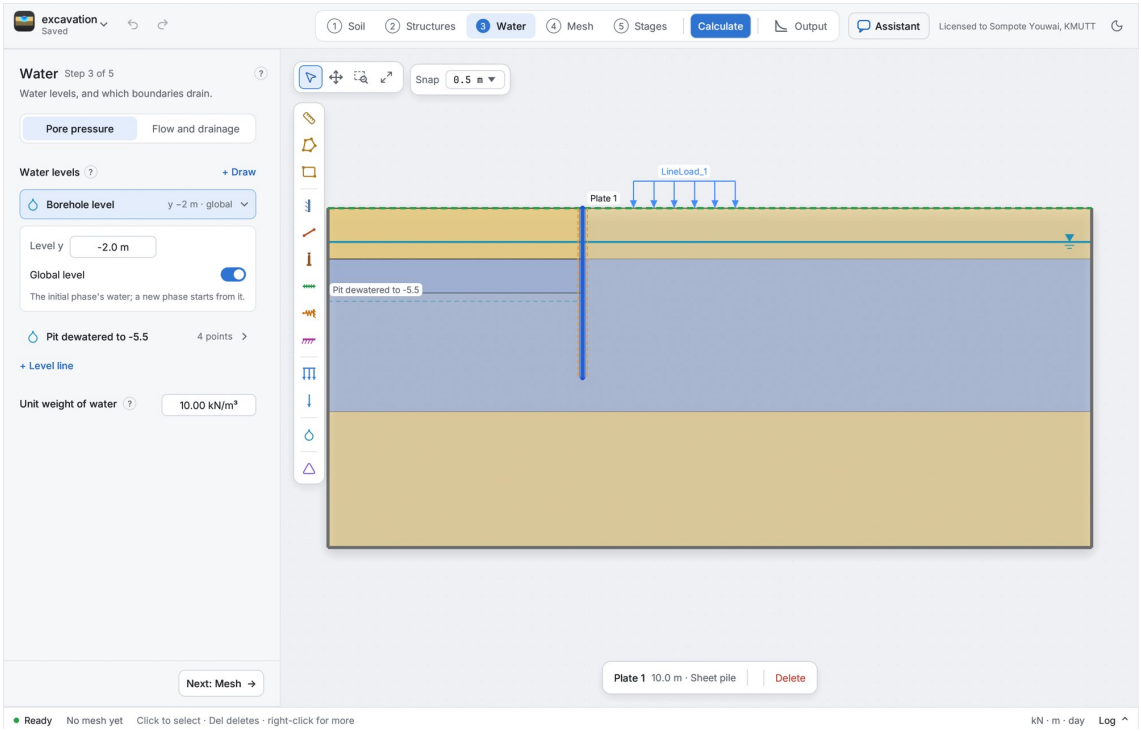


Figure 37. The Pore pressure tab: the global borehole level and a drawn level for the dewatered pit.

Water that stands above the ground presses on the surface as a load, so a reservoir against a slope, a flooded pit or a pond is modelled by drawing the level above the ground, and the K0 procedure counts the weight of that water. A soil that is **non-porous** has no pore pressure at all, which suits concrete or rock blocks.

A drawn level gives hydrostatic pressures under every point of the line. Next to a wall this is a jump in pore pressure across the wall and no head loss around its toe, which is on the safe side for the wall and too optimistic for uplift of the pit floor. Use steady flow (Section 11.4) when the flow around the toe matters.

11.3 Measured pore pressure profiles

Where the pore pressure is not hydrostatic, for example under-drained Bangkok clay or ground around deep wells, a water level can carry a measured profile, a table of pore pressure against elevation. The pressures then follow the table between its points and are hydrostatic beyond its ends, and they never exceed the hydrostatic value under the line, so lowering the line into a pit still dewaters the pit. Through MCP the profile is the option `pore_pressure_profile` of `add_water_level`:

```
m.tool("add_water_level", {
    "name": "Drawdown", "points": [[0, -2], [70, -2]],
    "pore_pressure_profile": [[-2, 0], [-12, 100], [-23, 0], [-60, 370]]})   # [level, kPa]
```

In the Sukhumvit excavation of Section 9.4 this profile halved the wall deflection compared with hydrostatic water, so ask for piezometer readings before you assume a hydrostatic profile in such ground.

11.4 Steady-state groundwater flow

With **Steady flow** chosen under Pore pressure in a phase, Tellus solves the groundwater flow through the porous soil that is in place and takes the pore pressures from it. The permeabilities kx and ky of each material control the pattern, plates and interfaces let no water through, and the sides of the model follow the **Flow and drainage** tab (Figure 38). An open side below the chosen water level has that level as its head, an open side above it is a seepage face where water may leave, and a closed side lets nothing through. Excavated surfaces and the free surface of drawn soil always drain. Above the water surface the permeability falls with suction, so the free surface is found by the calculation.

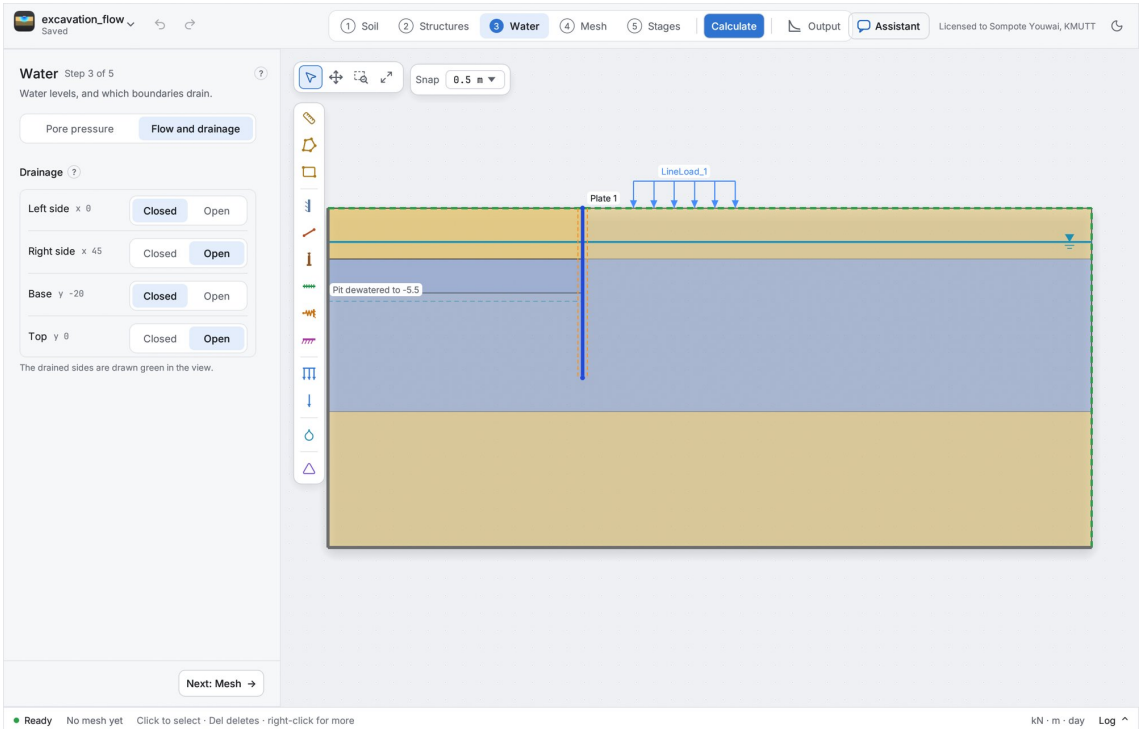


Figure 38. The Flow and drainage tab: each side of the soil open or closed. Here the outside boundary is open, and the axis of the pit and the base are closed.

Side	Default	Open it when
Left and right	Closed	The side is a source of water at the chosen level, for example the far field behind a wall. Keep an axis of symmetry closed
Base	Closed	A permeable layer below the model drains or feeds it
Top	Open	It is nearly always open: the ground surface drains

Figure 39 shows the head from a steady flow calculation in the last excavation phase of Chapter 4, with the outside boundary open at the level of -2 m and the pit floor draining. The head falls from about -2.3 m at 25 m behind the wall to -5.0 m at the pit floor, most of it around the wall toe, and the phase reports an inflow of 1.10 m³/day per metre run of wall. The permeabilities in this example are the default 1 m/day for every soil, so the inflow is an illustration and not a prediction for a real clay.

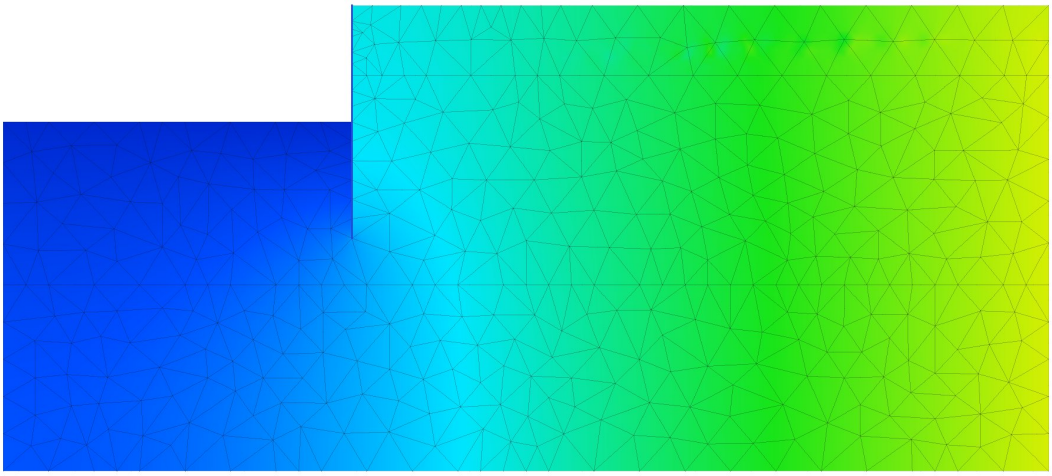


Figure 39. Groundwater head from steady flow into the dewatered pit, from blue (-5.5 m) to yellow-green (about -2 m). The head drops around the wall toe.

End of excavation (phase 3)	Phreatic levels	Steady flow
Largest wall movement into the pit	12.2 mm	9.1 mm
Largest bending moment in the wall	52.7 kNm/m	39.8 kNm/m
Strut force	64.4 kN/m	47.5 kN/m
Inflow to the pit	not calculated	1.10 m^3/day/m

The flow lowers the pore pressure behind the wall, because the head is lost on the way to the toe, so the wall carries less water pressure and its moment and the strut force drop by about a quarter in this example. On the pit side the upward flow raises the pore pressure under the floor, which reduces the passive resistance and matters for heave in fine sand. The flow is steady only. Time-dependent groundwater flow is not modelled, and consolidation phases handle the dissipation of excess pore pressure with time.

11.5 Drained and undrained soil, and consolidation

Drainage type	Strength input	Excess pore pressure	Use it for
Drained	c', φ'	None	Sand and gravel, and the long-term state of clay
Undrained (A)	c', φ' (effective)	Calculated from the soil model	Clay in the short term, when the model predicts the pore pressure well, and as the start of a consolidation analysis
Undrained (B)	su (entered as c, with φ = 0), effective stiffness	Calculated	Short-term stability with a measured undrained strength
Undrained (C)	su, undrained stiffness Eu and ν close to 0.5	None: a total stress analysis	A simple total stress check
Non-porous	As drained	No pore pressure at all	Concrete, rock blocks, structures modelled as soil

In a **Plastic** phase undrained soil keeps its water, so loading builds excess pore pressure. The switch **Undrained: ignore** of a phase treats all soil as drained in that phase, which is used for the long-term state or for a first loading stage that is long past. A **Consolidation** phase lets the excess pore pressure dissipate over its time in days, through the open sides of the Flow and drainage tab, and a load or soil that the phase adds is applied gradually over that time. By default

only the top drains, so open the base when the clay lies on sand, or the consolidation takes four times as long as it should.

Two points of the calibration (Section 9.3) apply here. The permeability stays constant during consolidation, so in very compressible soil the excess pore pressure dissipates faster than in programs that reduce the permeability with the void ratio. And a consolidation phase takes at most 250 time steps, so split a very long consolidation into several phases.

11.6 The mesh

Tellus divides the soil into 15-node triangles, which are accurate elements: each has 12 stress points and describes a displacement field of the fourth order, so a mesh of a few hundred elements already gives good deformations. The mesh is generated by **Calculate** when it is missing, or at once with **Generate mesh** in the Mesh step, and it is generated again after any change of the geometry. Two controls decide the element size: the global size, and a coarseness factor on each object.

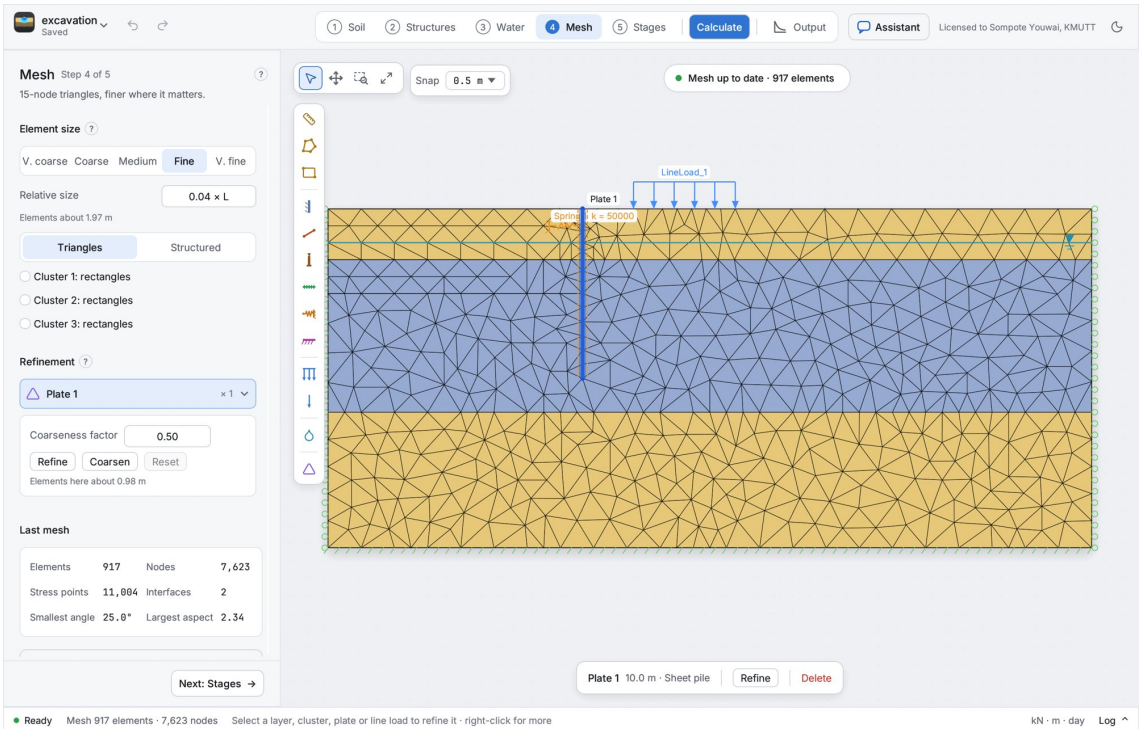


Figure 40. Step 4 Mesh: the global element size, triangles or a structured mesh, the refinement of the selected object, and the statistics of the last mesh.

Preset	Relative size	Elements in the excavation of Chapter 4
V. coarse	0.12 × L	
Coarse	0.08 × L	274
Medium	0.06 × L	
Fine	0.04 × L	917
V. fine	0.03 × L	
Typed value	0.025 × L	2158

The relative size is a fraction of the model size L, and the panel shows the resulting element size in metres under it. You can type any value in place of a preset.

11.7 Local refinement

Each soil layer, soil cluster, plate and line load has a **coarseness factor**, and the target element size in and next to the object is that factor times the global size. The default is 1 for soil and 0.5 for plates and loads, where the stresses concentrate, and the factor can range from about 0.03 to 8. Select an object in the picture or in the **Refinement** list and press **Refine** or **Coarsen**, which divide or multiply the factor by √2, so two presses halve or double the element size, or type the factor. **Reset** returns it to the default. Refined objects are drawn green and coarsened ones yellow. Away from a refined object the elements grow back to the global size within a few metres, and a coarseness factor changes the mesh only, never the model.

Refine here	Why
The toe of a wall and the corner of an excavation	Stresses and plastic points concentrate there
The edges of a footing or a load	The contact stress peaks at the edge
The toe of a slope and the expected slip surface	A Safety phase finds a sharper mechanism and a lower, more accurate factor
Thin layers, interfaces and geogrids	A thin layer needs at least one or two elements over its thickness
Around pile tips and anchors	The load enters the soil at a point

Coarsen the soil far from the structure, such as the deep layer under an excavation or the far field of an embankment, to keep the element count down. The time of a calculation grows faster than the number of elements, so a mesh that is fine where it matters and coarse elsewhere is the efficient one.

11.8 Structured meshes

Under the element size, **Triangles** and **Structured** choose the mesh type. A structured mesh uses 25-node rectangles on a grid with a line at every geometry line (Figure 41), and it can be chosen for the whole model or ticked for single rectangular clusters. Rectangles and triangles meet node to node, so the two can be mixed, and where a sloping line crosses a grid cell Tellus fills that cell with triangles. Every grid line runs across the whole region, so the long side of a cell is split until no rectangle is more than three times longer than wide. Structured meshes suit level layers, consolidation of a soil column and site response, where regular rows of elements are wanted, and they work with plates, interfaces, consolidation, flow, the advanced soil models and Safety phases.

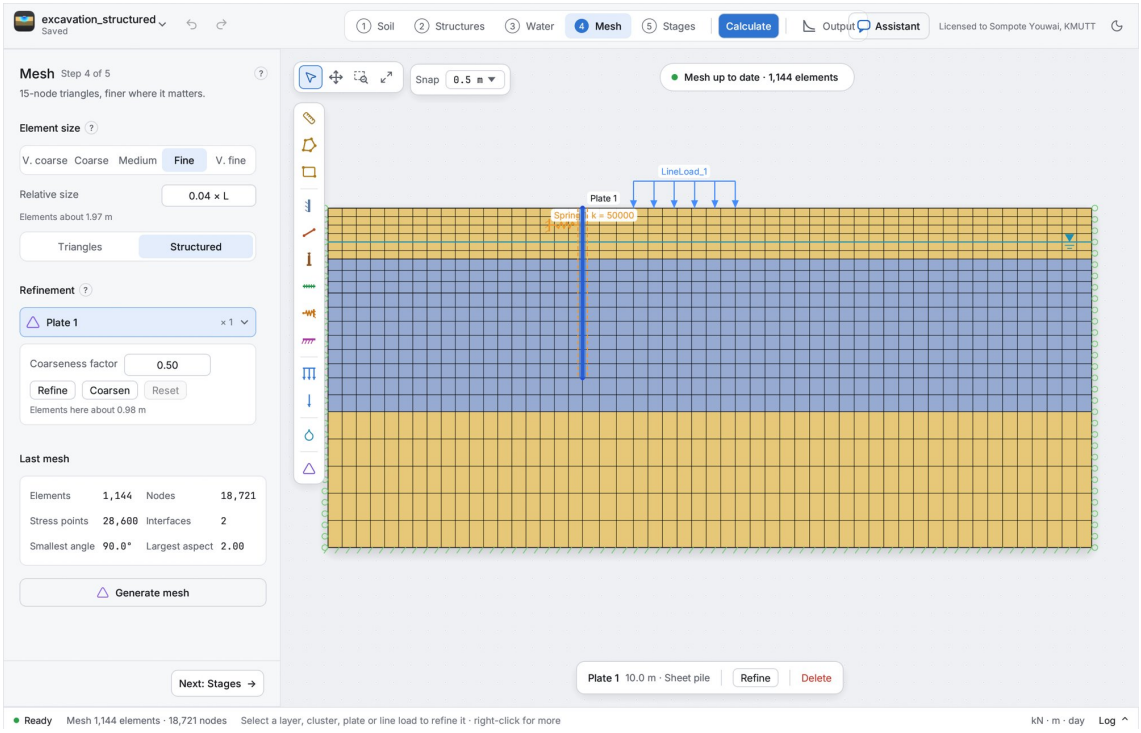


Figure 41. The same excavation with a structured mesh: 25-node rectangles on a grid through every geometry line.

11.9 Reading the mesh statistics

Value under Last mesh	Meaning	What to look for
Elements, Nodes	The size of the mesh. The number of equations is about twice the nodes	A few hundred to a few thousand elements for most models
Stress points	12 per triangle, 25 per rectangle	
Interfaces	The number of interface lines along plates	One per ticked side of a plate
Smallest angle	The sharpest corner of any triangle	Above about 20°. A very small angle means two geometry points lie too close together
Largest aspect	The longest side over the shortest of any element	Below about 3 to 4

A mesh with tens of thousands of elements from a simple model nearly always comes from two geometry points or lines that almost coincide, such as a cluster corner 1 mm from a layer boundary. Snap the points together and generate the mesh again.

11.10 How fine is fine enough

The only reliable answer is to run the model on two meshes and compare the result you care about. Table rows below give the excavation of Chapter 4 on three meshes. The wall moment changes by 0.6 % and the wall movement by 2 % between 274 and 2,158 elements, while the time grows twelve-fold, so the coarse mesh was already good enough for the wall forces of this problem.

Relative size	Elements	Nodes	Wall movement [mm]	Moment [kNm/m]	Strut force [kN/m]	Time [s]
0.08	274	2353	12.22	52.9	63.6	0.4

Relative size	Elements	Nodes	Wall movement [mm]	Moment [kNm/m]	Strut force [kN/m]	Time [s]
0.04	917	7623	12.19	52.7	64.4	1.6
0.025	2158	17689	12.00	52.6	64.5	4.5

Do not carry this conclusion to other results. Deformations and structural forces converge quickly with these elements, whereas a factor of safety from a phi-c reduction and a collapse load keep falling as the mesh is refined along the slip surface, by up to a tenth between a coarse and a fine mesh. Check a stability result on a refined mesh before you rely on it, and in a dynamic analysis choose the element size from the wavelength (Section 8.2).

11.11 Water and mesh through MCP

```
# water
m.tool("set_contour", {"x_min": 0, "x_max": 45, "water_level": -2.0})      # global level
m.tool("add_water_level", {"name": "Pit dewatered to -5.5",
                           "points": [[0, -5.5], [15, -5.5], [15.01, -2], [45, -2]]})
m.tool("set_drainage", {"x_min": "closed", "x_max": "open",
                        "y_min": "closed", "y_max": "open"})
m.tool("update_phase", {"phase": 2, "water_level": 0, "pore_pressure": "steady_flow"})
m.tool("update_soil_material", {"material": 1, "kx": 1e-4, "ky": 1e-4,
                                "drainage": "undrained_a"})
head = m.tool("get_results", {"phase": 3, "quantity": "head",
                              "points": [[14, -8], [16, -8]]})

# mesh
m.tool("set_model", {"model": {"mesh_size": 0.04}})          # global relative size
m.tool("set_model", {"model": {"structured_mesh": True}})    # rectangles
print(m.tool("generate_mesh"))                               # elements, nodes, quality
```

The coarseness factor of a single layer, cluster, plate or load is the field `coarseness` of that object in the model, which `get_model` shows and `set_model` changes. To an assistant you can simply say “refine the mesh around the wall toe and tell me how much the moment changes”.

12 Further options, good practice and troubleshooting

12.1 Updated mesh for large displacements

The switch **Updated mesh** in the settings of a plastic or consolidation phase (`updated_mesh` in the MCP tools) calculates the phase with large displacements. Equilibrium is then written on the deformed mesh, plates and piles turn with their chord, so an axial force acts with the lever arm of the deflection, and a geogrid stretches along its deformed line, so it carries load as a membrane. Use it when the ground moves by a good part of an element or a layer thickness, for example an embankment on very soft soil or a reinforced fill on a void. A phase that starts from an updated mesh phase uses it too. The option is verified on a compressed column and on a bending cantilever. In the source of 12 November 2026 a case in which plates and piles are installed in a mesh that has already deformed did not run, so check that combination on your model before you rely on it.

12.2 Calculating on a cloud GPU

A large or long calculation can run on another computer while the model stays on yours. On the other computer, `tellus --http 0.0.0.0:8787 --token <secret>` starts a calculation server, and an NVIDIA card with the cuDSS library speeds up the factorisation. On your computer, the MCP tool `set_compute` with the server's URL and token sends every calculation there, from the window, the assistant and `tellus --mcp` alike, and `set_compute local` goes back. The script `cloud/vast_gpu.sh` in the source rents a Vast.ai box, installs Tellus and sets it as the server, and `tools/build_linux.sh` builds the Linux program to upload. The calculation server needs no licence seat, while `tellus --mcp` on the box needs one.

12.3 Good practice

Topic	Advice
Model size	Keep the side boundaries at least three times the excavation depth or slope height away, and an embankment model three times its base width wide. Widen the model once and see that the result holds
Mesh	Start coarse to find mistakes quickly, then refine where the stresses change fast: wall toes, footing edges, slope toes. Repeat with a finer mesh and compare
Initial stresses	K0 procedure for level ground and level layers, gravity loading for slopes and sloping layers
Undrained soil	Undrained (A) uses effective strength and needs a model that predicts the pore pressure well. Undrained (B) uses su directly and is the safer choice for a short-term stability check
Phases	Model the real sequence. A wall wished in place before excavation, and each strut installed before the dig below it, as on site
Safety	A factor shown with ≥ has not reached failure in its steps. Look at the mechanism in the displacement picture and at the curve of ΣMsf against displacement
Dynamics	Check the signal's units, the time step against the record's, the element size against the wavelength, and the damping curve against the frequencies of interest
Checking	Open the deformed mesh of every phase. A wrong support, a missing interface or a load of the wrong sign shows there at once

12.4 Troubleshooting

What you see	Likely cause	What to do
“Tellus is not activated on this computer”	No licence token on this computer	Activate in the app or with tellus --activate (Section 2.3)
“All seats of this licence are in use”	The key is active on its full number of computers	Deactivate another computer, or ask for more seats
A load “lies outside the soil contour”	The model was made narrower than a load or structure	Move or delete the object, or widen the model
A plastic phase stops with the soil body collapsing	The load cannot be carried: real failure, or a missing support, too low a strength, a wrong water level	Read the load fraction reached and look at the deformed mesh of the last step
Later phases “not calculated”	The phase they start from did not finish	Fix that phase first
“Maximum number of steps reached”	A consolidation phase needs more than 250 steps	Split the time into several phases
A consolidation phase is very slow	Soft Soil Creep with many plastic points takes many iterations per step	Use a coarser mesh for a first run, or the GPU server
The assistant does nothing	No API key, or the provider's URL or model name is wrong	Open Assistant, Settings, and check the three entries
Tellus is missing from /mcp in Claude Code	The server was registered with a wrong path, or the session is old	Run claude mcp list, register again with the full path, start a new session
A dynamic run drifts or blows up	No drift correction, a time step larger than the record's, or far too coarse a mesh	Switch on drift correction, reduce the time step, refine the mesh

Appendix The MCP tools

The 37 tools of `tellus --mcp` in the source of 12 November 2026. Every tool describes its arguments to the assistant, so you do not need to remember them, and `get_model` with `set_model` reach any part of the model that has no tool of its own.

Group	Tools
Project	new_project, open_project, save_project, get_model, set_model
Soil	set_contour, set_soil_layers, add_soil_material, update_soil_material, add_soil_cluster
Structures and loads	add_plate, add_anchor, add_spring, add_geogrid, add_embedded_pile, add_load, add_fixity, set_boundary_conditions
Water	add_water_level, set_drainage
Phases	set_initial_phase, add_phase, update_phase
Dynamics	add_signal, set_monitors, get_dynamic
Calculation	validate, generate_mesh, calculate, set_compute, get_compute
Results	get_results, get_plate_forces, get_curve, show_picture, write_report
Exchange	export_plaxis_commands

Phase option (add_phase, update_phase)	Meaning
calc	plastic, consolidation, safety or dynamic
start_from	The phase it starts from, so phases can branch
time	Consolidation time in days
loads, plates, springs, anchors, geogrids, piles	What is active, and the load values
layers_on, clusters_on	Soil in place: false is excavated, true is placed
water_level, pore_pressure	The water level of the phase, phreatic or steady flow
ignore_undrained, updated_mesh	Drained behaviour in this phase, large displacements
vacuum	Vacuum preloading in a consolidation phase
dynamic	Duration, time step, boundaries, base motion and dynamic loads of a dynamic phase

Result quantity (get_results)	Meaning
utot, ux, uy	Displacements from the start, in m
phase_utot, phase_ux, phase_uy	Displacements of this phase
inc_utot, inc_ux, inc_uy	Displacements of the last step, which show a failure mechanism
sig_xx, sig_yy, sig_zz, sig_xy, p_mean, q	Effective stresses, mean stress and deviatoric stress, in kPa, compression negative
p_steady, p_excess, head	Steady and excess pore pressure, groundwater head